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e-Book | 2025

RF FILTER DIGEST

A Resource of Whitepapers and Articles

Highpass
Filters



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Band Reject
Filters



Bandpass
Filters



Introduction

With the introduction of new wireless technologies, the frequency spectrum is getting very crowded. To ensure these technologies can operate simultaneously without impacting each other, RF Filters play a very important role. RF Filters are crucial in optimizing signal transmission and reception, selectively allowing desired frequencies to pass through while attenuating unwanted signals.

everything RF has published the 2024 “The RF Filter Digest” eBook to act as a comprehensive resource for anyone looking to learn about the current-generation RF Filters. RF Filters not only enhance the performance of communication systems, but they also ensure efficient spectrum utilization while minimizing interference. This eBook discusses the challenges involved with 5G mmWave filtering, high-rejection LTCC filter performance, the role of MMIC filters in developing next-generation systems, and the impact of BAW filters on 5G applications. The eBook also includes an exciting section that discusses filter design in the Cadence AWR design environment.

The 2024 RF Filter Digest has been created in close collaboration with some of the leading companies in the RF Filter space. These include Qorvo, Mini-Circuits, Marki Microwave, Knowles and Modelithics.

Content Contributed by:



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The Surprising Ways MMIC Filters Are Enabling the Development of Next Generation Systems

Filters are ubiquitous throughout RF systems, yet they have not seen the same degree of advancements (e.g. size reduction and high frequency operation) that have been made to the rest of the RF signal chain. Current filter solutions consume a large volume of a system's overall footprint and are susceptible to process and lot-to-lot variation. Due to the custom nature of filter requirements, they are often designed in-house, consuming valuable engineering time and resources through multiple iterations of cascade analysis and subsequent design.

Until recently, MMIC has not been considered a viable technology for filter development due to misconceptions about the technology including development time, cost, suitability for custom solutions (due in part to the aforementioned time and cost), and the focus on traditional filter metrics, such as Q factor. In reality, MMIC design cycles are short due to fast wafer fabrication cycles and accurate first pass design success. Combined with the high-volume production capability of MMIC technology, this has the added benefit of reducing the cost of custom designs at volume.

Current RF market trends indicate that next generation systems demand significant reduction in component size. As a result, filter specifications are being reprioritized to ease loss and rejection requirements in exchange for size reduction. This reassessment of the trade-space has allowed MMIC planar filters to emerge as the ideal solution to these shifting market trends. Due to the precision of MMIC lithography, when compared against other planar technologies, MMIC filters have demonstrated unmatched scalability, repeatability and size reduction while enabling millimeter-wave designs. This whitepaper aims to discuss what Marki Microwave contends to be the key metrics of a modern competitive filter technology, how these are achieved through a GaAs MMIC process, and the capabilities of MMIC filters that have been demonstrated thus far. Together with the companion design tool that has been developed, this offers a fast, accurate and trustworthy way of developing filters without consuming large amounts of customer and vendor design resources.

Filter Idealizations

RF/Microwave filters are electronic circuits that allow only specific frequencies to pass through a system while rejecting unwanted signals outside of the desired passband. Filters are commonly used to provide channel/band selection and reject spurious tones,

improving system dynamic range. These designs are typically custom and specific to an application's frequency plan. Multiple filters are often needed at different points throughout a particular design.

The fundamental building block of filter designs is a resonator. An LC resonator

in a filter typically consists of an inductor and a capacitor in either a series or parallel configuration. These reactive components along with other circuit elements determine a filter's frequency response characteristics. In addition to the series and shunt LC resonators, planar designs can be implemented using distributed elements, such as spiral structures or tapered traces to achieve the desired inductance and capacitance.

In the context of planar filters, the layout of an LC resonator is implemented in a two-dimensional form on a flat substrate combining resonators with transmission lines.

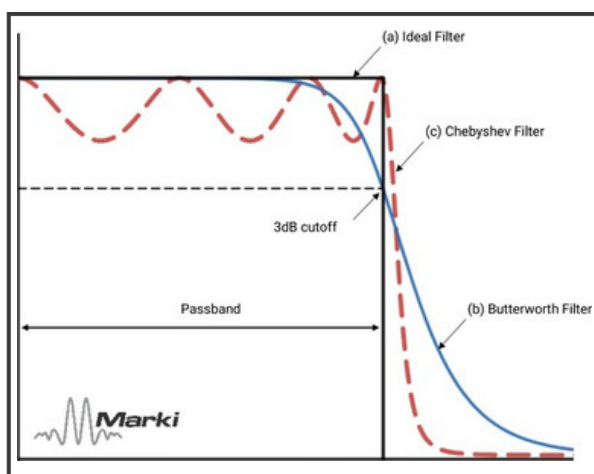


Figure 1. (a) Ideal filter frequency response; (b) Butterworth filter response; (c) Chebyshev filter response.

The Ideal (Brick-Wall) Filter

The ideal filter is a two-port device that would allow a desired range of frequencies to pass through a system with no loss and would immediately cut-off unwanted signals at the passband edge with an infinitely steep transition and out-of-band rejection as shown in Figure 1(a). Ideal filters are impossible to realize, even in digital signal processing.

Butterworth and Chebyshev Filters

The next level of filter idealization are the Butterworth and Chebyshev filter approximations. While the brick-wall response is not achievable, designers can

get closer to an ideal transition and cutoff by increasing the order of a filter. These filters are mathematically possible to achieve if you have perfect, lossless resonators. A Butterworth filter, as seen in Figure 1(b), features a maximally flat passband (low ripple) with a gradually sloping transition region at a rate of -20 dB/decade for a 1st order filter. Increasing filter order increases that steepness; a 2nd order filter would see a rate of -40 dB/decade, and a 3rd order filter would see -60 dB/decade, and so on. A Chebyshev filter, as shown in Figure 1(c), on the other hand can achieve a steeper transition region due to the allowable ripple in the passband and at the passband edges. Chebyshev filters are used when steeper roll-offs are critical for a design.

There are many other mathematical 'filter prototypes' that can be used as the starting point for a filter design, including Legendre, Bessel, Elliptic, and others without formal names. However a real, well-designed planar filter will not strictly follow any of the filter prototypes due to nonidealities in the physical implementation.

Real Resonator Filters

Filters with real resonators will have insertion loss and finite Q. Real resonators will introduce resistive and parasitic losses and the filter will have lower selectivity. The additional losses result in the widening of the filter bandwidth. When laid out and 3D simulated, crosstalk and other physical effects need to be considered as well.

Many different filter topologies exist; common topologies include: lowpass, highpass, bandpass, notch, and diplexer. The unique filter requirements of a system will dictate which filter topology is the preferred solution.

Filter Figures of Merit

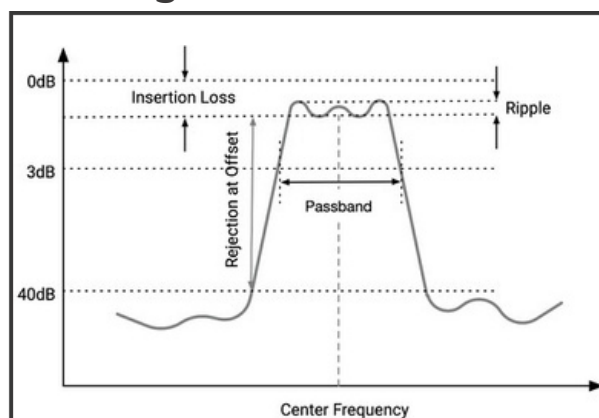


Figure 2. Key performance metrics of a microwave filter.

The goal of any filter design is to accommodate the design specification of a customer's system. Figure 2 displays the key performance metrics of a filter design. Typically, this will be the maximum insertion loss across the passband and the out of band rejection at some specified frequency. The resonator Q and filter order are key factors in defining a filter's center frequency insertion loss and rejection skirts.

Resonator Q

The Q, or quality factor, is defined as the ratio of the energy stored in a resonator to the energy.

$$Q = 2\pi \frac{\text{energy stored}}{\text{energy dissipated per cycle}}$$

Equation 1

A filter with a high Q will have a sharp resonant peak at the center frequency and a narrow bandwidth. Filters with a high Q factor are highly selective with respect to their passband.

Filter Order

The filter order, N , indicates the number of zeros or poles in a filter's transfer function. A higher order filter is achieved by increasing the number of reactive components (e.g. LC or other resonators) in the filter design. These

circuits are often designed using cascaded sections. A higher order filter typically features steeper roll-offs at the passband edges, narrower transitions between the passband and stopband and high stopband suppressions.

Insertion Loss

The filter order, N , indicates the number of zeros or poles in a filter's transfer. The center frequency insertion loss of a filter can be approximated by Equation 2, where $\sum_{i=1}^N g_i$ represents the sum of the resonator element values up to the Nth resonator, and Q_u is the unloaded Q factor.

$$\text{Loss}(f_c) = \frac{4.343 \cdot \sum_{i=1}^N g_i \cdot f_c}{\Delta f \cdot Q_u}$$

Equation 2

A filter with a high Q would demonstrate low center frequency insertion loss, while a higher order filter would present with higher loss, but steeper rejection slopes and higher selectivity. Additionally, a higher order filter would be a more complex, larger filter design. For this reason, the traditional approach to improving filter performance has been to focus on improving Q.

Filter Technologies

Common filter technologies in use today include acoustic wave filters, cavity filters, lumped element filters and planar filters. No single filter technology is best suited to fit all applications; each technology has its own tradeoff space. Cavity filters for example provide excellent out of band rejection, low loss and high power handling at microwave frequencies at the expense of being physically large designs that require hand tuning. Acoustic filters on the other hand offer excellent rejection in a small

form factor but are limited to frequencies below 8 GHz and with low power handling. Planar filters (e.g. MMIC, Thin Film, Laminate) fall somewhere between these technologies in size, but feature an excellent tradeoff for improved performance, size, cost and development time. Marki Microwave contends that MMIC filters are well suited to fit current demands for frequency and performance in a small form-factor. Table 1 provides a high-level view of what can be realized across different filter technologies, with 5 representing the strongest capabilities, and 1 representing the worst

A Reassessment of Performance Metrics

Technology	MMIC	Thin Film	Laminate	Cavity	Lumped
Size	5	3	2	1	2
Power	2	2	4	5	3
Performance					
< 10GHz	2	3	3	5	5
10 to 20GHz	3	3	4	5	2
20 to 50GHz	4	3	3	4	1
50 to 100GHz+	4	N/A	N/A	3	1
Q Factor	2	3	3	5	4
Practical Filter Order	2 (<9 th order)	3 (<11 th order)	4 (<15 th order)	5	3
Narrowband limit % BW	>8%	>6%	>5%	Any	Any
Transmission Zeros	3	2	2	5	4
Cost per design	1	3	5	3	5
Cost for multiple designs	4	2	5	1	5
Yield	5	3	3	3	4
ASP Scaling	5	3	4	2	4
Capable Frequencies	1 to 100+ GHz	1 to 40 GHz	1 to 40 GHz	0.1 to 40 GHz	DC to 20 GHz
Best-suited Frequencies	6 GHz to mmWave	5 to 20 GHz	5 to 20 GHz	1 to 20 GHz	DC to 6 GHz
Simulated v Measured Accuracy	5	4	3	3	3
Integration Capability	5	3	4	1	1

Table 1. Comparison of commonly used filter technologies.

As next generation systems aim to take advantage of wider bandwidths and support higher channel densities, RF market trends for filters are shifting to prioritize size-weight-and-power (SWaP), higher frequency designs, scalability and rapid custom filter development over preferences of low insertion loss and high out-of-band rejection.

For traditional filter technologies, this is problematic; as a filter's size decreases, there is a natural trade-off in Q and filter order. The implications of this are that the center frequency insertion loss takes a hit, passband edges are more rounded and the transition slopes are not as steep. However, if insertion loss requirements can ease slightly, MMIC filters are uniquely suited to become a viable solution for modern filter designs. With MMIC, the flexibility in filter topology and architecture provides more tools than just Q & filter order to play with.

In Marki Microwave's experience, filter specifications of modern and next gen applications appear to be less sensitive to insertion loss than they have been in the past. If designs are allowed to take a small insertion loss hit, then planar filter sizes can shrink significantly with MMIC. The goal of a filter design is to decrease size as much as possible without taking large Q penalties while still meeting the design spec: typically maximum passband loss and out-of-band rejection minimum at some offset frequency. Since lower Q is directly related to lower order filters and increased insertion loss, the trade-space in creating smaller filters is to find the balance between reduced filter size, rejection and insertion loss. Focusing specifically on planar filter technologies, when considering Table 1, MMIC filter designs offer the best in frequency capability, size, scaling, repeatability, integration capability, and agreement between simulation and measurement. While MMIC designs may have a lower Q compared to technologies such as laminate and thin film, the reprioritization of design specs means that the filter Q, while important, is no longer the most critical filter metric for many applications.

MMIC Filter Capabilities

High Frequency Designs

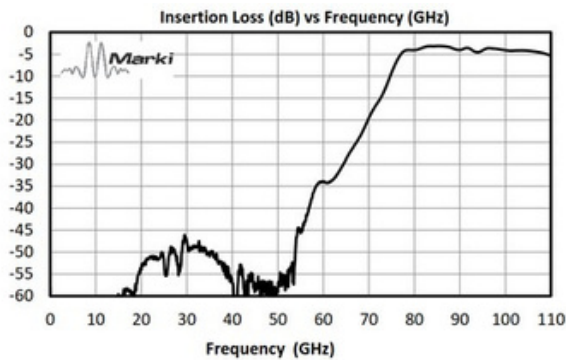


Figure 3. MMIC filter performance demonstrated to 110 GHz.

The precision and fine lines of MMIC lithography and the relatively thin substrate used enable high frequency filter designs that are not achievable with competing technologies. Combining unique design topologies and layout with high volume MMIC production processes, standard frequency capability has been demonstrated up to 110 GHz as shown in Figure 3.

Insertion Loss

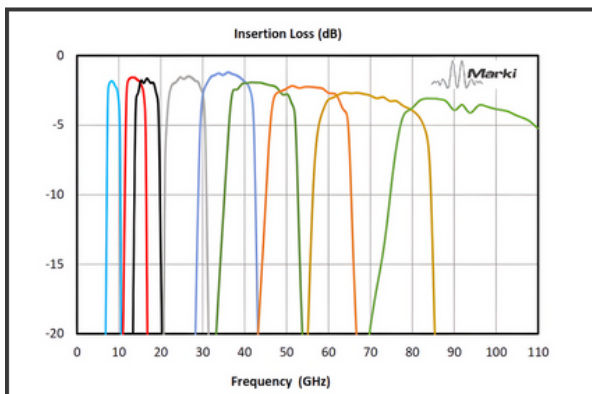


Figure 4. Measured passband insertion loss versus frequency for a selection of bandpass filter designs.

Typical passband losses for MMIC designs have so far been demonstrated between 1 and 5 dB, as shown in Figure 4, with lower frequency designs displaying superior losses. The move to higher frequencies and smaller filter sizes necessitates the use of thinner substrates. As a result, a filter takes a penalty in Q and thus a hit in insertion

loss, however, in exchange we gain frequency/bandwidth and more precise lines. The insertion loss of MMIC filters increases slowly at higher frequencies, whereas other technologies such as laminate and thin film tend to struggle in terms of insertion loss above 40 GHz.

Additionally, the insertion loss of a filter is also directly impacted by filter order and percent bandwidth as noted by Equation 2. While MMIC filter order is limited to below 9th order circuits, Marki Microwave has found that a filter with an order between 4 and 8 is sufficient to meet most rejection specs while maintaining an acceptable passband insertion loss.

Rejection In addition to the insertion loss across the passband, a typical filter spec is also determined by the rejection at a specified frequency outside the passband. For most cases, Marki Microwave has found that a 40 dBc minimum stopband rejection is achievable at 10% from the band edge. At higher frequencies, stopband rejection does take a slight hit due to slower roll-off, however, a 30 dBc minimum rejection above 40 GHz has still proven to be acceptable for most designs. **Return Loss**

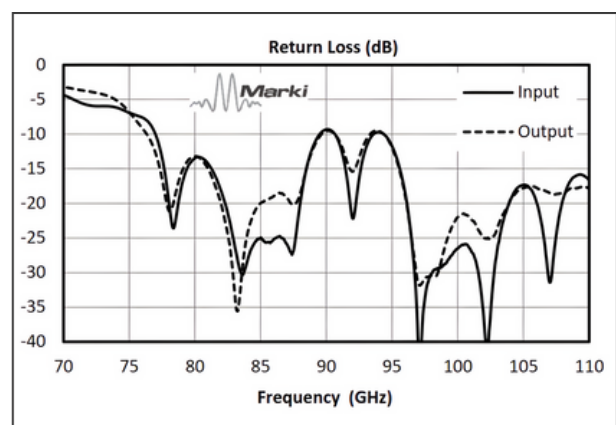


Figure 5. Measured return loss for a 78 GHz - 110 GHz MMIC bandpass filter.

For a standard reflective filter, a high (negative) return loss indicates that a majority of the input signal is being transmitted through the filter, ensuring efficient transfer and minimal signal loss. A high return loss indicates a well-matched filter that effectively distinguishes between the passband and stopband. MMICs have demonstrated measured passband return losses

better than 10 dB for filter designs above 100 GHz (Figure 5) as well as typical return losses in the 15 to 25 dB range at frequencies below 40 GHz.

Tolerance & Repeatability

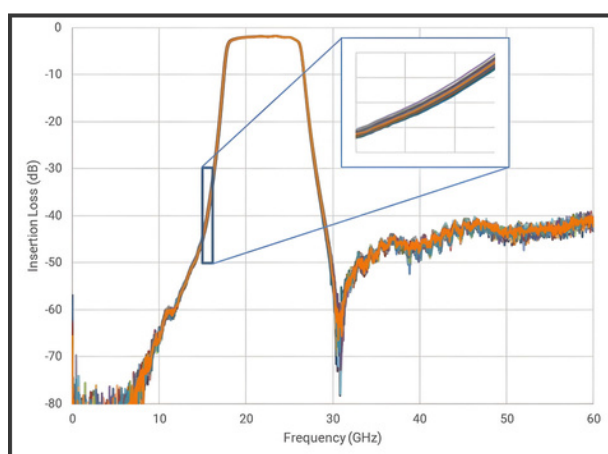


Figure 6. Overplot of a representative sample of 24 MMIC filters from a lot of over 8,000 filters demonstrating outstanding fabrication repeatability.

GaAs fabrication utilizes tightly controlled single crystals that can be produced in high volumes while maintaining repeatability. MMIC filter production is scalable from thousands to millions of units. The fine geometric features available on the process allow for the careful tuning and optimization of features such as transmission zeros via cross-coupling of resonators. These results have been achieved with near perfect match with simulation, enabling first pass design success and a short development cycle.

A tighter tolerance results in better repeatability from unit-to-unit and

wafer-to-wafer when compared against other technologies. While the design of a typical planar filter is similar among the different technologies (laminate, thin film, MMIC), the etching tolerance of a MMIC is unmatched. The typical etching tolerance of a laminate board is 25 μm whereas a typical MMIC is 0.25 μm . For the laminate filter, poor tolerances can result in dimensional variations that may detune the filter, lowering yield. As for thin film, the manufacturing processes are not as well controlled compared to ICs due to the less sophisticated, low volume tools and techniques employed resulting in variability from lot-to-lot.

Figure 6 displays an overplot of 24 randomly selected MMIC filters from a lot of over 8,000 filters demonstrating high repeatability with low variation in both passband frequency and loss.

MMIC Filter Application

Considerations

Package & Size

The heavy increase in channel count and total bandwidth demanded by next gen systems has driven the need for the substantial size reduction that can be achieved with MMIC filters when compared against other traditional planar filter techniques. The precision of MMIC lithography as well as the availability of extremely small, lumped element inductors and capacitors enables unique circuit innovation and significant size reduction of filter circuits. MMIC filters can be realized with die as small as 1.5 x 1.5 mm for high frequency designs. Additionally, a MMIC filter can be integrated directly with another MMIC block or co-packaged into a surface mount package or multi-chip module to further reduce a system's footprint.

Standard QFN vs Thin Film

Both the standard QFN and thin film

design.

High pass filters are typically AC coupled at the input therefore limited by a max RF power handling specification while low pass filters typically have a DC short in the through path with a max DC current capability. Filters with larger elements and wider traces (as seen in distributed element filters) enable higher power handling capabilities, however this comes with increased die dimensions.

Performance over Temperature

Aerospace and military applications require components to withstand extreme temperature ranges, with minimum and maximum operating temperature ranging from -55°C to 125°C. Over this temperature range, MMIC filters have shown very low performance variation as seen in Figure 8. However, the magnitude in variation between insertion loss and return loss is dependent on filter design topology. Lumped element filters experience greater temperature variation compared to distributed element filters at very high frequencies. As element sizes shrink, the tolerance becomes a significant factor as any slight variation will cause the resonant frequency to shift. With temperature changes, material expansion and shrinking will also cause shift in frequency. Increased temperature will generally increase insertion loss while decreasing temperature will yield

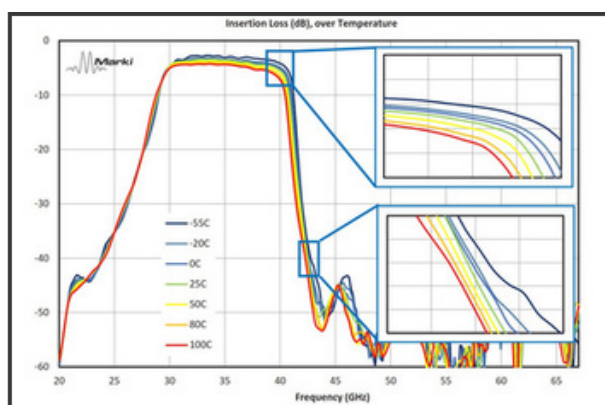


Figure 8. MMIC filter insertion loss variation over temperature.

improved insertion loss performance. In general, lower percent bandwidth filters will also be more sensitive to temperature change.

The Future of MMIC Filters

Filter specifications tend to be custom to their projects. They may be planned for at the initial design phase or potentially towards the end to clean up unexpected spurious signals. Regardless, it is critical that custom filters can be developed quickly and accurately. The future of MMIC filter design is to heavily minimize the design burden for both the customer and design engineer.

Marki Microwave recently demonstrated a new GaAs MMIC filter design tool, Filter Prodigy, based on AI-generated finite element models and Marki's ever-growing library of existing filter designs. Using the tool, a customer will be able to select the filter type and design parameters (center frequency, %BW, etc.) of their project, and the tool will almost instantly provide accurate s-parameters and die dimensions. On the backend, Filter Prodigy manipulates existing filter layouts in real time and produces a 3D model, again almost instantaneously, that meets the customer spec, nearly eliminating the bulk of the engineering design work that would typically go into laying out a filter. As the library of filters grows and new topologies are added, the tool will only become more accurate.

Figure 9 demonstrates a 10 GHz filter design at varying percent bandwidths simulated using the Filter Prodigy tool. As expected by theory, as percent bandwidth increases, insertion loss improves while the rejection slopes become less steep. Even then, the insertion loss between a narrowband and wideband filter differs by approximately dB. The customer application will

determine whether this loss is feasible, but regardless, these design questions can be determined quickly using this tool.

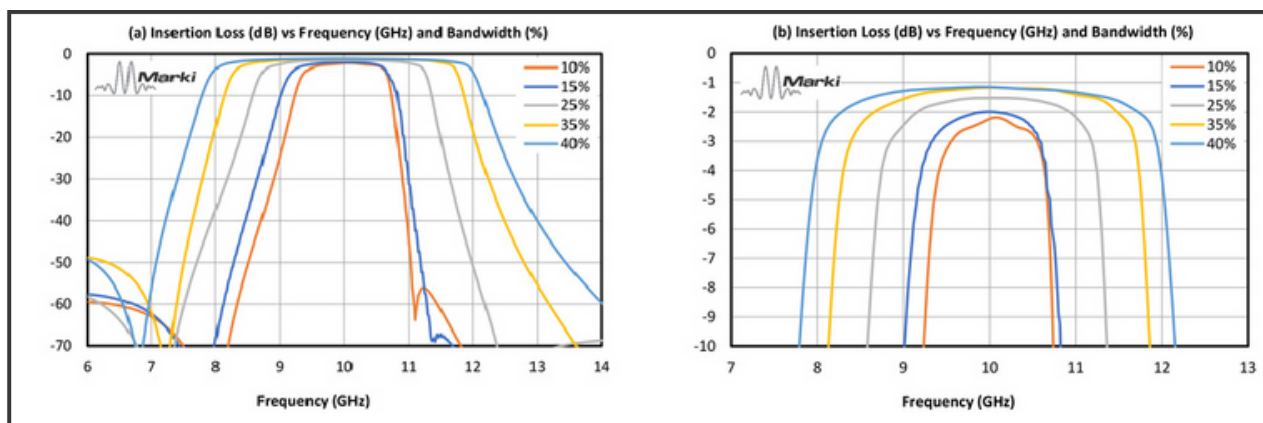


Figure 9. Filter Prodigy simulation of Insertion Loss vs Frequency at varying %BW

For a customer, this is a fast, accurate and trustworthy, iterative tool. Using the provided s-parameters from each design, they themselves can quickly simulate their filter, check and modify their requirements against their system requirements and repeat the process before bringing Marki into the discussion. Once a design is reached that satisfies their requirements, Marki can verify their design for accuracy. If filters need to be packaged, some additional 3D modeling may be required, but the bulk of the work has already been completed. This, combined with the capabilities of MMIC as discussed, allows first-pass design success and results in shorter design cycles that are not achievable with other filter technologies.

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PERFORMANCE THAT SEEMS UNREAL

Introducing Prodigy™ Filter Designer

Go from Ordinary to Extraordinary RF/Microwave Development

RF systems have unique frequency plans and utilize RF filters to select bands of interest and reject unwanted signals. This often drives the need for custom filters and an iterative system optimization process, resulting in costly engineering time and longer design cycles. **Generate instantaneous 3D FEM quality MMIC filter designs with Prodigy™, the new automated filter design tool from Marki Microwave.** Powered by advanced FEM modeling and machine learning techniques, Prodigy puts MMIC filter design into the customer's hands.

Start designing at www.markimicrowave.com/technical-resources/tools/filter-prodigy

The Trusted Leader When Performance Matters



- ✦ Generate accurate, custom MMIC filter designs with instant access to s2p files
- ✦ Enables fast, iterative system optimization
- ✦ Cost-effective & streamlined delivery of products



RF Filtering for 5G millimeter wave applications

This whitepaper will discuss some of the areas of Fifth Generation (5G) communications that will be implemented in the millimeter wave (mmWave) region of the RF spectrum and will guide the reader through some of the options available to implementing RF filtering at these frequencies. In the first part of this whitepaper we will do a short review of mmWave frequencies and their applications, both prior to the advent of 5G and with planned 5G systems. In the second part we look at filtering technologies available for use at mmWave frequencies.

About millimeter-wave frequencies

The millimeter-wave part of the electromagnetic spectrum is at the high end of the microwave region (which spans ~300 MHz to 300 GHz) and is usually taken to mean frequencies from ~30 GHz to 300 GHz and wavelengths in the range of 1mm to 1cm. The ITU designation for the band spanning 30 GHz to 300 GHz is EHF (Extremely High Frequency).

One of the characteristics that makes this part of the spectrum so interesting to communication applications is that the frequencies are that much higher than those historically used in wireless so the available spectrum in the mmWave potentially opens up 200x the spectrum available below 3 GHz.

From a dramatic increase in available bandwidth comes an increase in achievable data-rates, making these frequencies extremely interesting to teams around the world working on 5G communications.

However, with this increase in frequency comes a set of challenges. The free space path loss (FSPL) equation provides a way to calculate signal attenuation through free space or a vacuum. It tells us that signal loss increases as the square of distance and the square of frequency. The higher the frequency the more the signal will degrade over distance, limiting these frequencies to short range and/or high-power communications.

$$FSPL = \left(\frac{4\pi d}{\lambda}\right)^2 \quad \begin{array}{l} d = \text{distance between the antennas} \\ \lambda = \text{signal wavelength} \end{array}$$

In addition to losses in free space, radio signals are attenuated both by atmospheric constituents and by physical objects (trees, buildings) that they encounter. mmWave frequencies have proven particularly susceptible to these factors. For example, there is an oxygen absorption band in the range of 60GHz, signal loss due to rain and humidity is a known engineering challenge throughout the mmWave range, and attenuation through materials

Band	Symbol	Frequency range (lower limit exclusive, upper limit inclusive)	Corresponding metric subdivision
4	VLF	3 to 30 kHz	Myriametric waves
5	LF	30 to 300 kHz	Kilometric waves
6	MF	300 to 3000 kHz	Hectometric waves
7	HF	3 to 30 MHz	Decametric waves
8	VHF	30 to 300 MHz	Metric waves
9	UHF	300 to 3000 MHz	Decimetric waves
10	SHF	3 to 30 GHz	Centimetric waves
11	EHF	30 to 300 GHz	Millimetric waves
12		300 to 3000 GHz	Decimillimetric waves

Table 1. ITU Band Designations

like concrete or brick is increased for high frequencies.

With that said though these challenges are not insurmountable and even before 5G began to build interest a wide range of applications have utilized mmWave frequencies.

Current uses of mmwave technology

Communications

Microwaves (of which mmWaves are a subset) have been widely used in point-to-point communications. An example of point to point would be in a backhaul link where microwave frequencies are used to transmit data in a straight line of site link between a cellular base station and a base station controller. Point to point at mmWave frequencies is attractive because the reduced wavelength allows conveniently-sized antennas to focus them into narrow beams, which can be aimed directly at a receiving antenna. This prevents nearby microwave equipment that use the same frequency from interfering with each other, allowing the deployment of more links in the same area. In the US, 36 GHz to 40 GHz is licensed for high-speed microwave data links, and the unlicensed 60 GHz band can be used in short range data connections.

In the US, 36 GHz to 40 GHz is licensed for high-speed microwave data links, and the unlicensed 60 GHz band can be used in short range data connections. In the 60GHz unlicensed band another application is WiGig, or IEEE 802.11ad, intended to support high bandwidth gaming and 4k video streaming over WiFi. The new standard adds a new 60 GHz frequency band alongside the current 2.4 and 5 GHz offerings.

Military

Military applications of Microwave and mmWave have a long heritage.

Table 2 shows some example military applications across frequencies ranging from 2 GHz to 40 GHz. Typically, as frequencies increase the applications move to higher resolution applications.

Automotive

mmWave frequencies also find application in radar based Advanced Driver Assistance Systems (ADAS), helping drivers control vehicles and to assist in automated functions. These systems often use both short range (at 24 GHz) and long range (at 77 GHz)

Band	Range	Example Military Applications
S-Band	2-4 GHz	Weather Radar. Surface Ship Radar. AWACS systems. Communications Satellites. EW applications.
C-Band	4-8 GHz	Weather Radar. Radar Altimeters. Communications Satellites. EW applications.
X-Band	9-12 GHz	Fire Control Radar. Communications Satellites. EW applications.
Ku-Band	12-18 GHz	Fire Control Radar. Communications Satellites. EW applications.
K-Band	18-27 GHz	Inter Satellite Communication and Ranging.
Ka-Band	27-40 GHz	High Resolution, Close Range Targeting Radar.

Table 2. Some examples of Military Applications at Microwave Frequencies.

radar to scan the environment around the car. The high frequency, and thus short wavelengths, involved is a key advantage for mmWaves in this application. A radar system operating at 77 GHz will have a wavelength of approximately 3.8mm. As a consequence, the size of system components such as the antennas required to process signals is reduced. An additional benefit to using such short wavelengths is the increased levels of accuracy, since such a system will have

the ability to detect movements that areas small as a fraction of a millimeter

Mmwave applied to 5G Frequencies of interest to the 5G community

Earlier we talked about how the use of frequencies up in the mmWave makes large amounts of additional bandwidth available, which in turn permits an increase in transmission speed. Fifth Generation (5G) communication systems being planned to enable a hundred-fold increase in user data-rates – with this need for huge amounts of bandwidth in 5G, mmWave has become extremely interesting to the 5G community.

We also talked about some of the challenges involved in using mmWave frequencies, such as range limitations brought about by propagation losses. However recent advances in mmWave systems have made these frequencies a commercially attractive prospect and have turned some of the perceived disadvantages into key enablers in system architecture. For example, short transmission paths and increased propagation loss allow for spectrum re-use. And the established ability of mmWave in point to point communications to be tightly focused into beams allows signal strength to be directed exactly where it needs to go, reducing interference, and also allowing multiple beams to be combined, resulting in increased range.

Following the World Radio Communication Conference in 2015 (WRC-15), the ITU released a resolution addressing ‘possible additional allocations to the mobile services on a primary basis in portion(s) of the frequency range between 24.25 and 86 GHz for the future development of International Mobile Telecommunications

for 2020 and beyond’ including the following list of frequencies: 24.25-27.5GHz, 31.8-33.4GHz, 37-40.5GHz, 40.5-42.5GHz, 45.5-50.2GHz, 50.4-52.6GHz, 66-76GHz, 81-86GHz.

Since then, a group of frequencies have come forward as the main candidates for mmWave applications in 5G: 28 GHz, 38 GHz, and 72 GHz. These frequencies have been selected because they have the following advantages:

- Reduced Oxygen absorption rates compared with other mmWave frequencies.
- Shown to perform well in multipath environments, allowing them to be used in non-line-of-sight (NLOS) communications
- Demonstrated to perform well with directional antennas, beam forming and beam tracking.

Potential uses for mmWave in 5G

To include 28 GHz conceptually under mmWave seems to be an industry wide convenience – technically (according to the ITU) 28 GHz would in fact fall in the centimeter-wave range. In the context of 5G mmWave is now being used to refer to the licensed bands such as 28 GHz (US) and 38 GHz (Korea) with the addition for unlicensed spectrum up at 60 GHz.

A detailed discussion of the potential use cases is described in the 2017 paper ‘Where, When, and How mmWave is Used in 5G and Beyond’. The applications discussed in that paper are:

28 GHz Outdoor Backhaul for Fixed Wireless Access

The idea behind Fixed Wireless Access is that high bandwidth connections into the home and business can be provided by

mmWave radio. Think about replacing cable and fiber with radio. The 2018 US commercial launch will be at 28GHz, and there is talk of trials in Japan beginning in 2018 at 60GHz. In addition, there is some discussion of placing access points in buses, for example.

60 GHz Indoor Access with Mobile Edge Computing

The paper discusses the use of 60GHz in locations such as airports in an attempt to support data intensive applications such as augmented reality (AR).

mmWave Mesh Networks for μ -RAN

This scenario involves the deployment of mmWave base stations as an overlay to existing LTE cells in dense urban environments to provide sufficient bandwidth. It is suggested that a 6 sector mmWave gateway would provide backhaul to a number of mmWave small cells in a mesh network approach.

mmWave based V2V/V2X for Automated Driving

In probably the most remote use case described in the paper, mmWave networking is used to support driverless cars in Vehicle to Vehicle (V2V) and Vehicle to Everything (V2X) applications.

At this point we can see that mmWave frequencies are going to be an integral part of the changes that 5G will bring about over the next few years. The different use cases that mmWave technologies enable makes up a diverse set of applications – each bring their own set of challenges when it comes to implementing systems in a robust and reliable way.

Something we know here at Knowles Precision Devices is that, just as we have seen filtering playing a key role in some of the more common uses of mmWave frequencies, filtering will continue to be a

major design component in radio access technology in 5G. Next we will take a look at mmWave filter technology options.

Filters in mm-wave applications

Here we provide an overview of some of the common technologies that are both in use in industry today and are generally available from more than one manufacturer. This last point is a factor to consider when looking at some of the more experimental alternatives– our own approach is to provide a solid foundation based on known implementations that are proven to be reliable.

On chip or off chip?

When we look at filtering options for mmWave RF systems, an early choice comes from choosing between on-chip (integrated into the RFIC for example) and off-chip (filtering outside the RFIC with surface mount components or connectorized solutions).

At these frequencies, for on-chip solutions the common semiconductor technologies are Silicon (Si) and Gallium Arsenide (GaAs). It is often desirable to use on-chip solutions because of not only the compactness of the circuits, but also the robustness of the on-chip filters since the manufacturing tolerances are those of the semiconductor manufacturing process. Integrating an on-chip filter with other devices to form a system-on-chip (SoC) solution, as is commonly done below 6GHz, can dramatically reduce the physical size of a wireless system. However, the reduction in dimensions of devices working in the mmWave compared to devices at 3GHz can make interconnect between devices challenging and to-date on-chip implementations have seen limitations when it comes to quality factor (Q),

losses and Noise Figure (NF). High isolation and low loss can be a challenge for on-chip filtering, and when a selfinterfering TX is in the same band as the RX, lumped filtering is often not practical for on chip solutions and an off-chip duplexer is usually required.

Challenges with implementing sufficiently highperformance filter structures on-chip arise from various factors, including the physical characteristics of the semiconductor material and the cost of implementation. In Gallium Nitride (GaN) for example, the aim is to make the circuits as thin as possible to encourage the heat dissipation necessary for the high powers that GaN is capable of. However, the Q of a filter structure is proportional to the thickness of the dielectric substrate, so in GaN this means an advantage of the material (high power) works in opposition to building filters with high Q. In addition, the space required to implement a filter structure in GaN takes up valuable chip real estate – area on the wafer than can be devoted to much more lucrative active systems such as amplifiers. At the time of writing, building on-chip filter structures with high enough Q to serve in a front-end application has proven impractical.

Our survey then will focus on off-chip filtering technologies, but it is worth keeping some of the advantages of on-chip (reduced size, repeatable and robust manufacturing) in mind as we go forward.

A broad view of filter technology

A good place to start our survey is by looking at the frequency ranges covered by different off-chip band-pass filter technologies.

We can look at seven common off-chip approaches to building filters:

Crystal Filters make use of a quartz crystal as the resonant element. The high Q of a quartz resonator makes for a very steep band-pass. These filters are usually implemented at IF frequencies in the range of 10 MHz and Q factors fall in the range of 10,000 to 100,000. SAW/BAW (Surface Acoustic Wave and Bulk Acoustic Wave). Acoustic filters cover a range of frequencies up to 6 GHz and offer a good performance/cost tradeoff, making them the dominant off chip filter approach in mobile devices today.

Ceramic filters cover a range of ~100 MHz to ~8 GHz. They offer similar performance to discrete lumped element inductor-capacitor (LC) designs but can be implemented in small form factor surface mount packages. Performance and package thickness can be a limiting factor when comparing ceramic filters with SAW/BAW.

Lumped Element. Discrete LC approaches provide a low-cost approach to implement a filter, but the attainable Q factors are limited in such devices. Discrete lumped element filters are usually used around the 30 MHz to 300 MHz range but can in principle be built for applications up to 40 GHz. At mmWave frequencies though discrete lumped element filters are very hard to implement because of the dimensional limitations imposed by the frequency, since the filter elements must be much smaller than the wavelength of the transmission lines.

Cavity filters are a common approach in the 40 MHz to 960 MHz frequency range and can offer high selectivity under high power. They can achieve good performance but are physically large, and usually only seen in infrastructure applications, such as for additional filtering at a cell site.

Planar filters are manufactured using a

thin-film process, and depending on the filter topology, can offer high Q and a reasonable approach to achieving performance in a small footprint when compared with discrete lumped element designs. In a thin film Lumped Element approach, the filter's transmission lines are printed in various configurations, depending on the required performance and filter elements are realized through discrete resistive, capacitive, and inductive elements. Planar Distributed Element filters rely on carefully distributed transmission lines to create resonant structures and can be designed to tighter tolerances than a lumped element filter. Distributed Element designs are more practical than Lumped Element designs at increased frequencies temperature stable ceramics to provide frequency stable performance over temperature

Waveguide filters are characterized by high power handling capability, leading to their wide adoption in radar applications and low loss given that the waveguide itself is a low loss medium.

Figure 1 summarizes the frequency ranges covered by these filter technologies. Looking at the plot the main choices for filtering in mmWave applications (e.g. ~30GHz and up) appear to be Cavity, Planar Thin Film and Waveguide. All three approaches bring advantages of their own and are based on established, readily available technologies. Let's take a look at some of the performance characteristics that are important in filtering applications, both in general and to 5G communications.

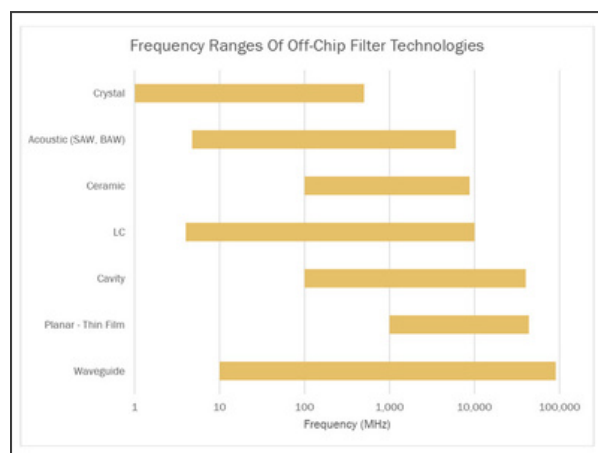


Figure 1. Common filter technologies and their frequency ranges

Filter Specifications To Consider

Percent BW: Where Bandwidth is the width of the passband of a bandpass filter and is expressed as the frequency difference between lower and upper 3 dB points, Percent Bandwidth is a common relative merit that compares bandwidth with carrier frequency. Commonly calculated as $3\text{dBW}/(\text{Center Frequency})$. Percent Bandwidth is likely to be an important consideration in 5G mmWave applications. One of the desirable attributes of the mmWave spectrum is the ability to access large amounts of bandwidth (and) hence increased data rates). To have a radio access system bandwidth limited because of available filter technology is not desirable.

Q factor: The quality factor (Q) of a resonator is a measure of the Selectivity of a resonator and is expressed as the ratio of stored versus lost energy per oscillation cycle. Overall losses through a resonator increase as Q factor drops and will increase more rapidly with frequency for lower values of resonator Q. As a result, the edges of the passband become more rounded and the bandwidth narrows as the Q decreases.

Selectivity is another way of talking about Q factor, and a measurement of the capability of the filter to pass or reject specific frequencies relative to the center frequency of the filter. Selectivity is typically stated as the loss through a filter that occurs at some specified distance from the center frequency. A filter with high selectivity exhibits high slope in the transition from pass to stop – Selectivity is crucial in environments where adjacent channels are close together and high selectivity enables designers to make good use of available bandwidth.

Insertion Loss: Loss of signal power through the filter. Important to consider on the Tx side because power is a system cost driver, and on the Rx side because loss impacts the overall Noise Figure of the receiver.

Temp Stability: Not often specified on filter datasheets the ability of the filter to hold its specifications (not just center frequency but percent BW and Selectivity) over temperature is an important factor to consider in systems that want to make good use of available bandwidth. The near term mmWave applications in 5G, such as FWA, will entail small scale systems (e.g. pole mounted) in exposed environments just as with today's Small Cells. Such deployment scenarios have raised the bar when it comes to a front-ends ability to withstand both extremes in temperature and temperature variation. Further, overall size reduction in systems lead to densely populated boards o heat from the surrounding components can impact the stability of the filter.

Power Handling: The ability of the filter to withstand large amounts of transmit RF power, this is mostly a concern in traditional macro-cell applications below 3GHz. In mmWave 5G applications the

transmit power will be spread over the individual elements in phased array antenna systems, and where the transmit range at these frequencies is reduced, filters do not need to withstand as much power.

Package Type and Package Size. These factors look at how the filter is physically implemented. Is the filter contained in a somewhat bulky connectorized package, or can it be implemented in a compact surface mount or flip-chip configuration? How consistent are the interconnects at mmWave frequencies – that is, how does the packaging impact the amount of post assembly tuning that a system will need to undergo before shipment? Component size will be a key enabling (or restricting) factor in practical mmWave front end designs. In phased array antennas the elements must be sufficiently close together to avoid generating grating lobes – and half wavelength spacing for mmWave frequencies amounts to a few millimeters. The resulting compact arrays will need to find a way to integrate the necessary filtering and depending where in the architecture RF filtering is deployed, space for filtering will come at a premium and compact packaging will be desirable.

Weighing up available filter technologies

Viewing the available technologies at mmWave, we can begin to make some general statements about what approaches are likely to be useful in a 5G system. An approach to doing this is to look at the factors identified above in Table 3.

When we look at some of the driving factors that will influence 5G mmWave front end designs, Planar Thin Film implementations emerge as a desirable approach from the standpoint of size,

cost and acceptable performance.

Characteristic	Cavity	Planar Thin Film	Waveguide
Frequency Range	Good	Better	Best
Percent BW	Best	Good	Good
Q Factor	Good	Good	Best
Selectivity	Good	Good	Best
Insertion Loss	Good	Better	Best
Temp Stability	Good	Best	Good
Power Handling	Good	Better	Best
Package Type and Package Size	Good	Best	Poor
Cost	Good	Best	Poor

Table 3. Factors influencing off-chip mmWave filter technology choice

Filtering in 5G applications will be just as important at mmWave frequencies as it is below 3GHz, but with a new set of challenges brought about by the available filtering technology and the characteristics of mmWave systems.

5G systems will require filters with high percent bandwidth, good selectivity, and excellent temperature stability in a . compact package but will not require the highest power handling capability.

Based on these criteria, the Planar Thin Film filter implementation becomes a desirable approach. Successfully implementing filters using Thin Film manufacturing requires the right choice of substrate (to ensure suitably high Q) and sufficiently high thin film tolerances (to ensure repeatable performance at the scales involved in mmWave applications).

Approaching this design challenge will be the subject of a future whitepaper

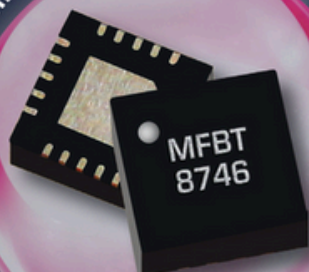
Conclusion

Frequencies in the mmWave spectrum will play a key role in 5G communications. RF technology that was developed around existing mmWave applications have evolved to encompass the needs of 5G wireless access.

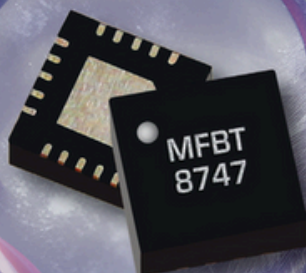
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PERFORMANCE THAT SEEMS UNREAL

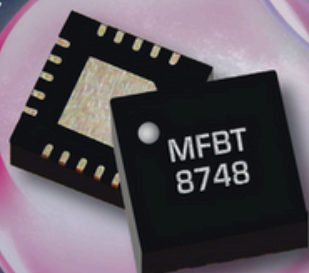
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A New Generation of 5G Filter Technology

5G implementation is accelerating worldwide, creating new challenges for wireless technology in mobile devices. The drive to deliver higher data rates with 5G has resulted in an enormous increase in RF complexity: the use of new higher-frequency bands, increasing complexity of modulation schemes, increasing number of antennas, carrier aggregation (CA) and E-UTRA New Radio Dual Connectivity (EN-DC) are all being used extensively to increase bandwidth. These trends increase the probability of interference in mobile devices, to mitigate this, advances in RF filtering technology are required. Space available for the RF front end (RFFE) is shrinking, which means filters must also become smaller and more highly integrated. If that wasn't enough, filters also need to handle higher power to maintain adequate design margins and maximize handset operating range.

Bulk acoustic wave (BAW) filters are evolving to address these challenges, along with other approaches such as antenna routing and antenna tuning. BAW technology is advancing toward handling higher frequencies up to 7 GHz – the upper limit of 5G frequency range 1 (FR1). BAW filters provide low insertion loss that helps maintain RF output power and steep skirts to avoid interference, which makes them extremely efficient at dissipating heat at high power levels. A new generation of BAW technology, called micro-BAW (μ BAW), saves board space by making filters smaller and easier to integrate into RFFE modules that support 5G requirements.

5G Filtering Challenges

Table 1 illustrates how RF complexity in mobile infrastructure and devices has increased with each generation of wireless technology. 5G adds yet another level of complexity, intensifying the challenges for mobile device design engineers. RFFEs in 5G devices must support many more RF paths, higher power output and much greater bandwidth, all while occupying less space – minimizing link losses and efficiently removing heat are vital. Advances in filter technology are critical to achieving all these goals.

Features	LTE Advanced			5G
	2015	2017	2019	2020
Number of total RF paths	14	22	>100	>300
Number of filter-band paths	7	12	18	>25
Maximum number of filters combined at one node for carrier aggregation	2	6	8	15
Critical isolation specifications required to meet standard	2	36	51	74
Max bandwidth of transmit signal	20 MHz	40 MHz	60 MHz	100 MHz
Power class 2 (3 dB higher power)	No	No	Yes	Yes
LNA integration	No	No	Yes	Yes
Envelope tracking or average power tracking (ET or APT)	APT	ET	ET	ET & APT
Incremental RF PCB size change	Baseline	+25%	+50%	-35%

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Table 1. Band and feature developments over time.

Power Handling and Heat Dissipation

Rising RF power requirements have increased overall system heat. To maintain handset reliability and performance, filters must be able to handle higher power levels and efficiently dissipate the additional heat.

RF power output in mobile devices has increased due to requirements such as Power Class 2, a specification introduced by 3GPP to improve 2.5 GHz TDD-LTE coverage worldwide. Power Class 2 raises power output at the handset antenna by 3 dB, as shown in Figure 1, to increase uplink (UL) range. This specification is also being used to support new higher frequency 5G bands, such as n77, n78 and n79.

Furthermore, handset manufacturers require RFFE components that handle higher power in order to maintain adequate design margins as system losses increase due to growing RF complexity. In today's mobile and infrastructure devices, filters must handle power levels as high as 33 dBm or more. By effectively dissipating the heat caused by these higher input power levels, filters can maintain performance and avoid lifetime degradation.

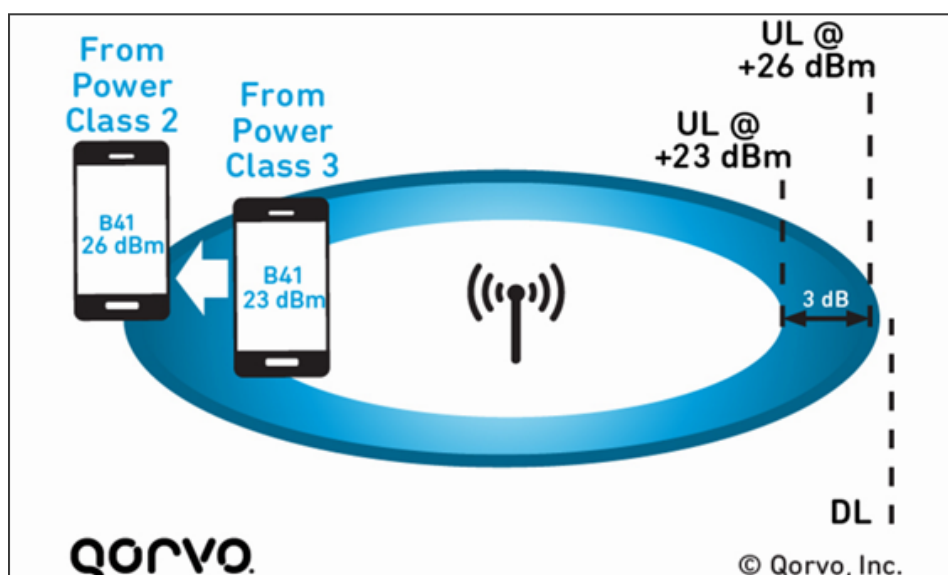


Figure 1. Power Class 2 increases handset uplink power and range compared to Power Class 3.

As shown in Figure 2, the structure of solidly mounted resonator (SMR) BAW filters removes this heat much more efficiently than other filter technologies, like film bulk acoustic resonator (FBAR). SMR BAW uses an acoustic reflector comprised of solid layers below the piezoelectric resonator. These solid layers provide a direct link to the silicon substrate below, allowing the heat generated within the filter to efficiently move away from the piezo resonator through the reflector layers to the substrate. In contrast, FBAR devices use an air cavity as the acoustic reflector. Because air is a poor heat conductor, heat must travel around the cavity rather than through it. Therefore, SMR devices are more efficient at moving heat into the silicon substrate, as shown in Figure 3. Increased system heat only has a minimal effect on SMR BAW filter and system performance, in comparison to FBAR, because of these structural differences.

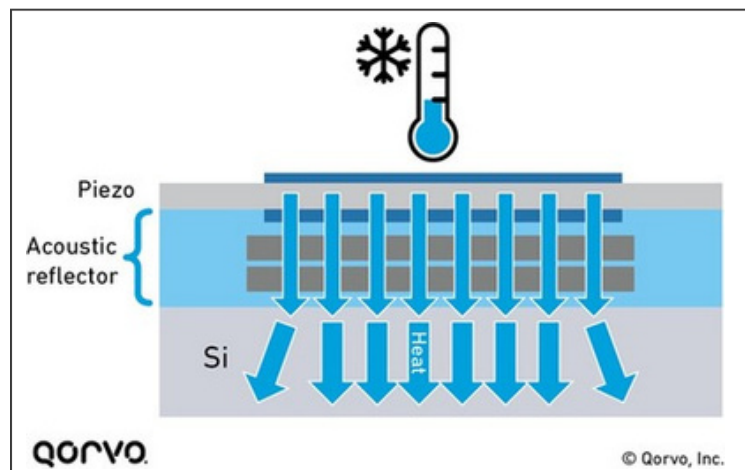


Figure 2. SMR BAW power and heat handling

Heat Dissipation at Higher 5G Frequencies

The advantages of SMR BAW over FBAR are even more pronounced at the higher frequencies used for 5G. Power handling in filters becomes more challenging at these frequencies due to the smaller resonator size. However, with SMR BAW, the reflector layers also become thinner, which further improves heat extraction (Figure 3). In contrast, with FBAR, the membrane above the air cavity becomes thinner, reducing its ability to move heat away from the resonator. As a result, with SMR BAW the temperature rises 20°C per Watt of transmit power, compared to 70°C rise per watt with FBAR. Less temperature variation means better performance, which enables SMR BAW filters to meet system requirements for insertion loss and out-of-band (OOB) attenuation under high-power and temperature conditions.

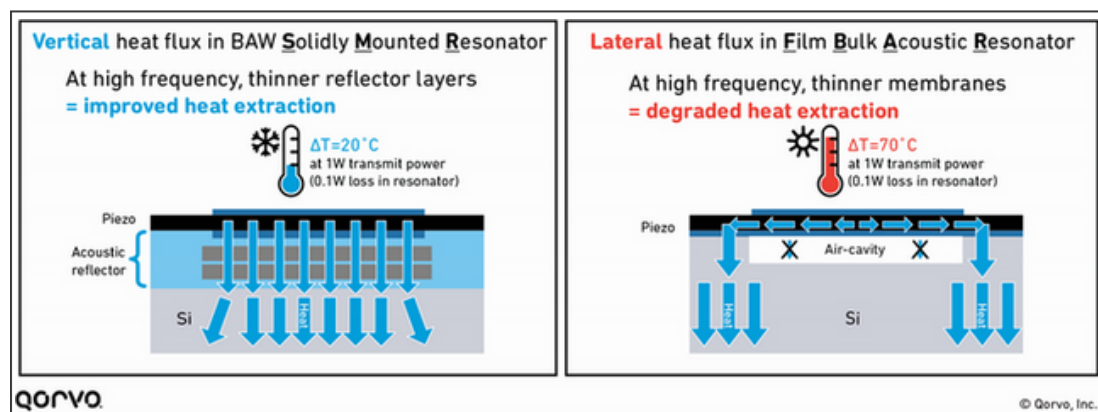


Figure 3. SMR BAW versus FBAR filter power and heat handling

SMR BAW filters also have high quality (Q) factors due to their low insertion loss and steep skirts, which also helps to reduce dissipated power and associated thermal issues.

Insertion Losses

As Table 1 shows, the number of RF and filter band paths has increased dramatically in LTE-Advanced Pro and 5G smartphones. These additional paths increase insertion loss as signals pass through the system. Any increase in insertion loss reduces RF output, especially in the higher 5G FR1 ranges above 3 GHz. PA output can be increased to

compensate for these incurred losses, but it comes at the cost of higher system current consumption. A better way to mitigate the system link budget increase is to use lower-loss switches and filters, integrate individual components into modules and reduce PC board line lengths. New SMR BAW generations offer reduced insertion loss to help design engineers meet link budget specifications in today's 5G devices.

Small Form Factor

As illustrated in Figure 4, PC board area in 5G smartphones is shrinking due to several factors. Phone manufacturers are increasing battery size to support 5G and other new features. To support broad range frequencies and new wireless standards including Wi-Fi, low-band, mid-band, high-band, ultra-high band and millimeter wave, more antennas are also being added. Therefore, system designers need the RFFE to occupy less area on the PCB. As a result, this drives semiconductor manufacturers to make filters and other components smaller and integrated into increasingly dense, complex RFFE modules.

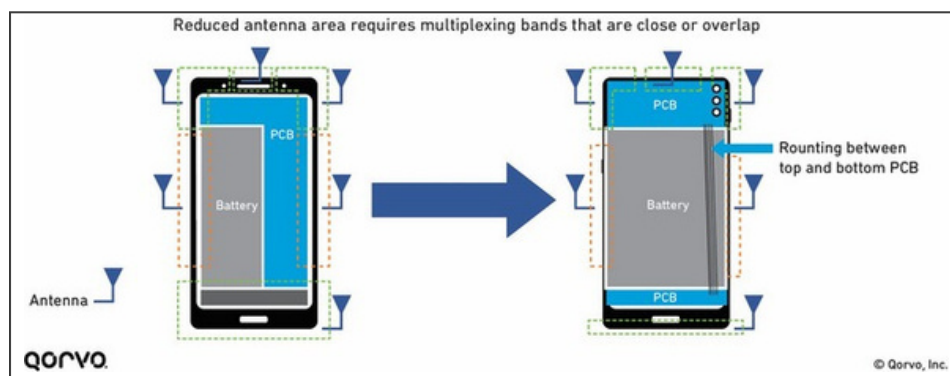


Figure 4. Larger batteries in 5G phones reduce the available RF related PCB space.

The development of μ BAW technology is a key step toward addressing these challenges in 5G smartphones. Like previous BAW generations, μ BAW uses wafer level packaging (WLP) with Cu-Sn-pillars. μ BAW reduces the die peripheral area by placing the device input and output terminals on top of the WLP-roof, overlapping with the active area. As shown in Figure 5, μ BAW shrinks the vias needed to connect the input and output pads to the BAW filter. This also means the distance between the active resonators can be further reduced when compared to previous BAW generations.

Out-of-Band Attenuation with BAW Multiplexers

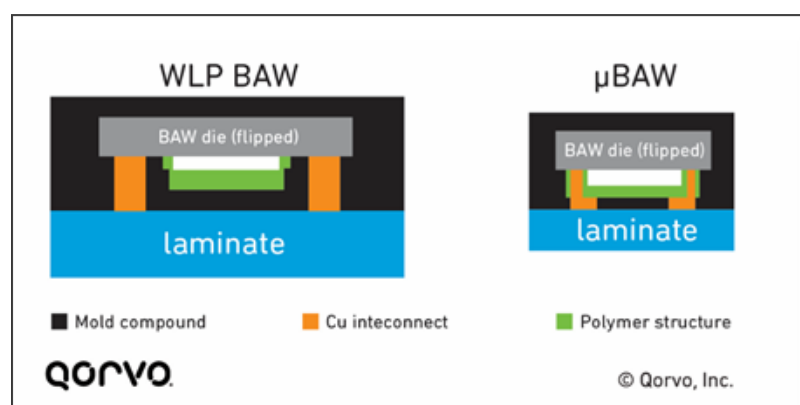


Figure 5. Comparing previous WLP BAW generations with μ BAW

In the 5G ecosystem, attenuating adjacent signals is becoming more difficult as technologies such as CA are used to increase device data rates. CA, first introduced with LTE Advanced, enables mobile network operators (MNOs) to maximize the use of their existing mobile spectrum and bandwidth by combining multiple component carriers into one data channel. By increasing uplink and downlink data rates, CA enhances network performance and enables MNOs to ensure a high-quality user experience.

5G allows many more carrier combinations than LTE, which increases the out-of-band (OOB) attenuation challenges. Enabling CA requires simultaneous communication on multiple carriers, which may be different frequency bands. The system must include mechanisms to help achieve isolation between these RF paths. BAW multiplexers achieve the required cross-isolation between the aggregated RF pathways, allowing communication to occur simultaneously on all the aggregated carriers while attenuating OOB signals for each pathway. BAW filters are well suited to attenuating OOB signals thanks to their steep skirts.

Advancements in filter technology were critical to enabling the adoption of CA with LTE Advanced Pro, and are equally critical for 5G. Filter material advancements in coupling, aluminum nitride enhancement and material doping were important milestones to achieving high attenuation, low return loss and the cross isolation required for enabling CA.

Advances such as μ BAW allow for the development of more complex integrated multiplexers, as shown in Figure 6. These highly integrated multiplexers reduce board space requirements and have the added benefit of reducing losses by eliminating the need for additional matching elements.

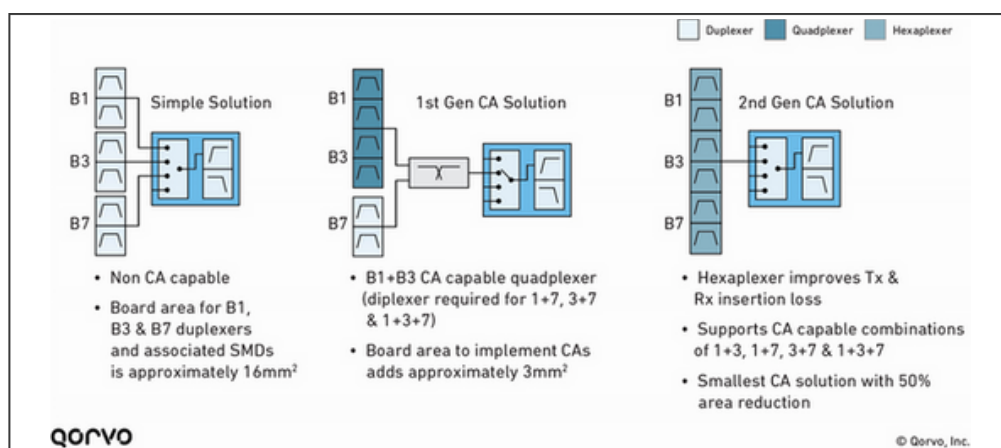


Figure 6. Tackling CA with advanced BAW multiplexer filter design

Conclusion

SMR BAW filter technology has been critical to enabling successive waves of wireless technology and is playing equally important roles in 5G mobile devices. New advances in BAW technology are mitigating thermal issues, reducing board space requirements, enabling greater integration, supporting complex CA requirements and handling the higher frequencies used for 5G. Future BAW technology developments will continue to provide system designers with the tools to further enhance our mobile devices and networks.

Optimizing High-Rejection LTCC Filter Performance in Co-Planar Waveguide Implementations

Mini-Circuits' BFHK-series of high-rejection LTCC filters have been characterized with stopband rejection floors on the order of 90 dB and higher with a combination of size, reliability, and cost currently unmatched by other filter technologies with comparable performance. Performance in the customer's system, however, can vary depending on the specific implementation. The unique design of these devices features coaxial input and output pins on the bottom surface of the structure requiring blind vias to the conductive layer of a stripline circuit board. While many PCB manufacturers have developed the capability to build surface-mount assemblies with blind vias reliably, some designers still prefer to use coplanar waveguide (CPW) wiring boards where the contact between the conductive trace and the device ports is exposed on the top layer. In addition to avoiding any concerns about blind vias, in certain use cases, CPW allows soldering of other surface-mount components in shunt or series to the signal trace as well as fine-tuning of the trace width and characteristic impedance for optimal matching conditions. To this end, Mini-Circuits has developed the BFHKL series of CPW-compatible filters by on a pick-and-place-ready platform, consisting of a sub-assembly with the LTCC component and an interposer substrate, which converts the coaxial launch of the LTCC into a CPW interphase.

This paper describes the physical differences between stripline and coplanar waveguide implementations of Mini-Circuits' high-rejection LTCC filters and related effects on performance. Channelization is proposed as an effective technique to achieve comparable performance to stripline implementations in CPW environments. Real test data from a leading customer's evaluation of the new BFHKL series of these filters on interposer boards in a channelized housing is then presented as proof of concept.

PCB Layouts in Stripline vs. Coplanar Waveguide Implementations

Because the land pattern of the BFHK-series filter package features conductive plating across the bottom surface as shown in Figure 1, simply soldering the unit to the exposed trace of a CPW board would cause functional issues due to shorting between the conductive metallizations on the PCB and the bottom surface of the filter.

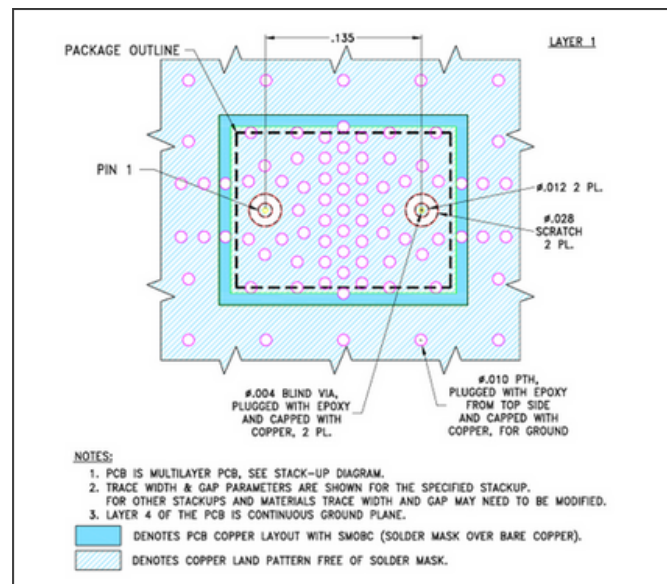


Figure 1: Stripline PCB layout for a BFHK-series high-rejection LTCC filter

Mini-Circuits developed the BFHKI series of filters incorporating a novel interposer board to allow drop-in use on CPW launches to facilitate integration on the user's end (Figure 2). The LTCC filter is pre-mounted on a stripline base designed for coaxial mating between the top-layer metallizations on the customer's PCB and the filter input and output ports as shown in Figure 3.

The interposer board enables surface mount assembly of Mini-Circuits' high-rejection LTCC filters on coplanar waveguide substrate, but as with everything in the RF world, it comes with caveats and requires due consideration in the design and assembly process.

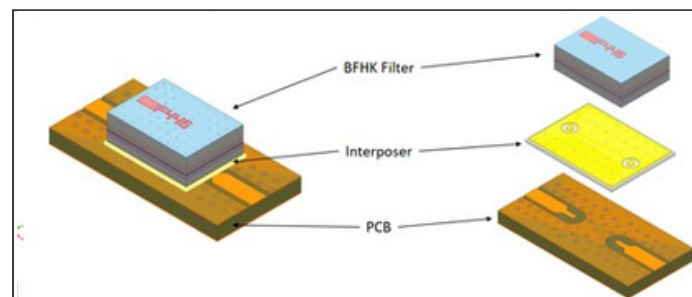


Figure 2: BFHKL-series LTCC filters feature an interposer between the filter and the customer's PCB, allowing easy use with top-layer transmission line.

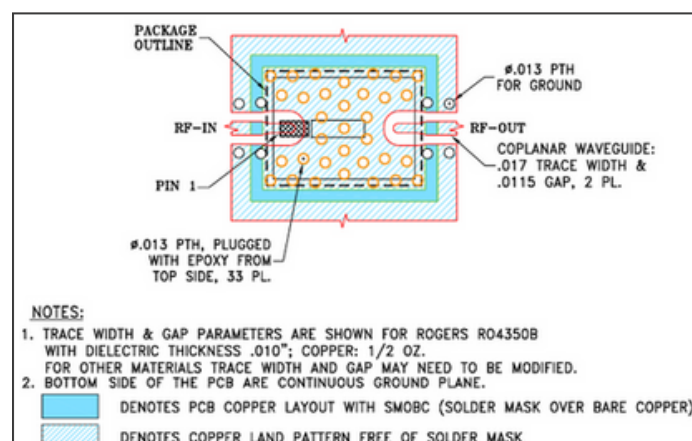


Figure 3: Suggested PCB layout for BFHKL-series LTCC high-rejection band pass filter on coplanar waveguide launches.

Effects on Performance

The specified rejection performance of BFHK-series filters is characterized on stripline test boards where the line to and from the filter is shielded from cross-coupling, buried in the PCB stackup. By contrast, in a CPW implementation, the exposed launch from the PCB to the device creates leakage, which affects rejection performance. For illustration, consider Figure 4. The filter response for BFHK-8501+ is shown on the left exhibiting a rejection floor of about 100 dB. The response for BFHKL-8501+, the same filter mounted on the interposer, exhibits typical rejection of about 70 dB in the lower stopband and 50 dB in the upper stopband.

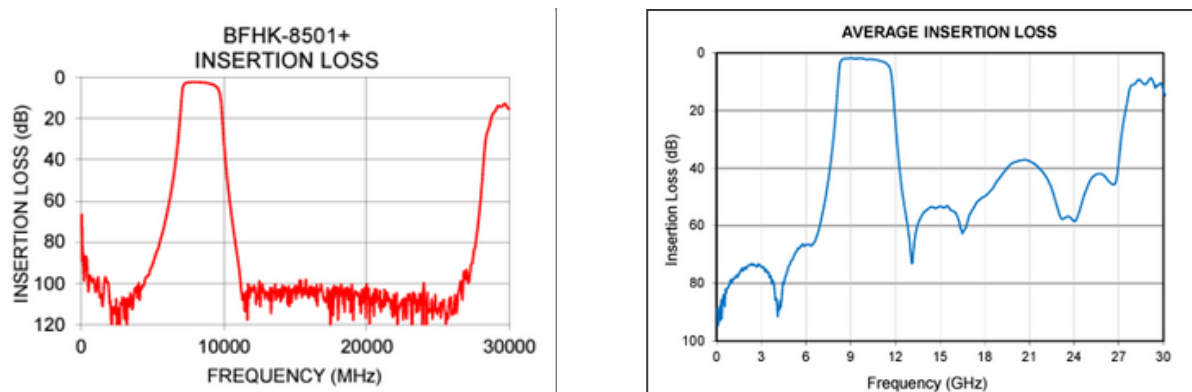


Figure 4: Characteristic response of BFHK-8501+ (left) vs. BFHKL-8501+ (right).

Note that in the test fixture for the BFHK-series model, the filter is mounted directly to stripline substrate, while the BFHKL-series model is characterized with the interposer mounted on an open CPW test board. It's important to qualify that the variation in rejection performance between the two models is a function of the quality of the launch from the PCB to the filter rather than an intrinsic property of the filter itself.

An ideal solution would replicate the insulation provided by the top layer of a stripline PCB in a CPW environment. Fortunately, this can be achieved by making use of the LTCC's conformal metallic coating and industry-standard channelization techniques, yielding a response similar to that of the BFHK-series mounted directly on stripline. What follows will present data measured by a leading customer in the test and measurement field evaluating the performance of Mini-Circuits' BFHKL-series filters on CPW substrate in channelized housings. The results will demonstrate performance parity between the baseline case of a filter mounted directly to stripline and the CPW use case with the interposer board and channelization.

The Test Setup

The customer used a 20 mil test board with four positions for BFHKL filters of identical layout using PC 2.92mm connectors. Input and output lines were grounded CPW types. Ground vias were made through the whole PCB stackup. Additionally, there was a TRL Calibration Board not shown here, which allowed de-embedding of the measurements excluding effects from PC 2.92mm connector and CPW line to the BFHKL unit. The test board and PCB layout are shown in Figure 5:

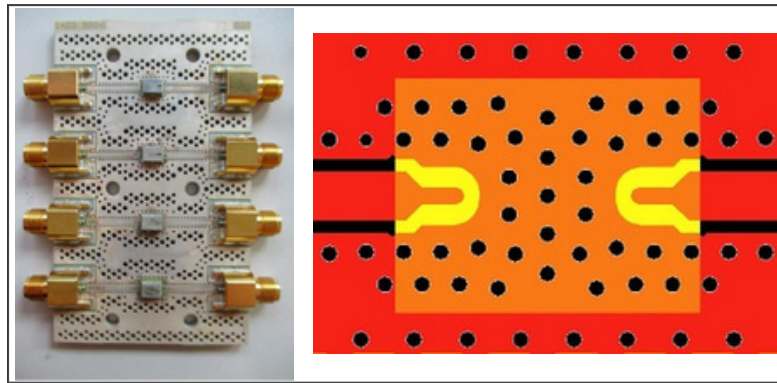


Figure 5: Test board and PCB layout for the customer's measurement of BFHKI filters.

Additionally, as an experiment, a conductive die attach adhesive (Ablebond 84-1LMI) was applied around the interposer except at contacts with RF lines. The adhesive application is pictured in Figure 6:

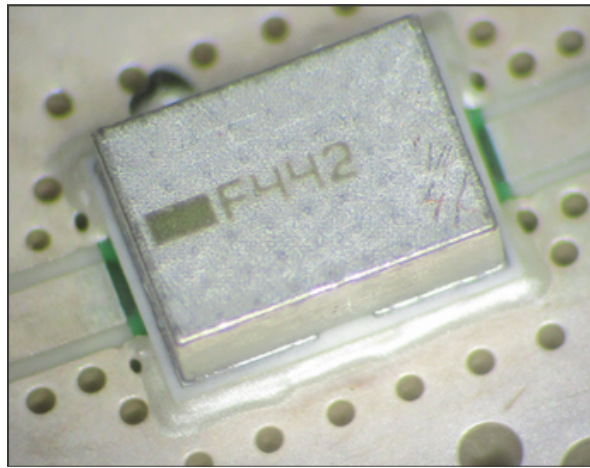


Figure 6: Conductive die attach adhesive was applied around the edge of the interposer board as a measure to maximize rejection.

Channelization with a Silver-Plated Conducting Cover

To test the filters in a channelized environment, a silver-plated conducting cover was used. The cover (pictured in Figure 7) contained single chambers for each of the four BFHKI filter positions on the test board. Note that the dimensions of the chambers were not optimized for the BFHKI filters specifically because the cover was built for use with devices of various sizes.

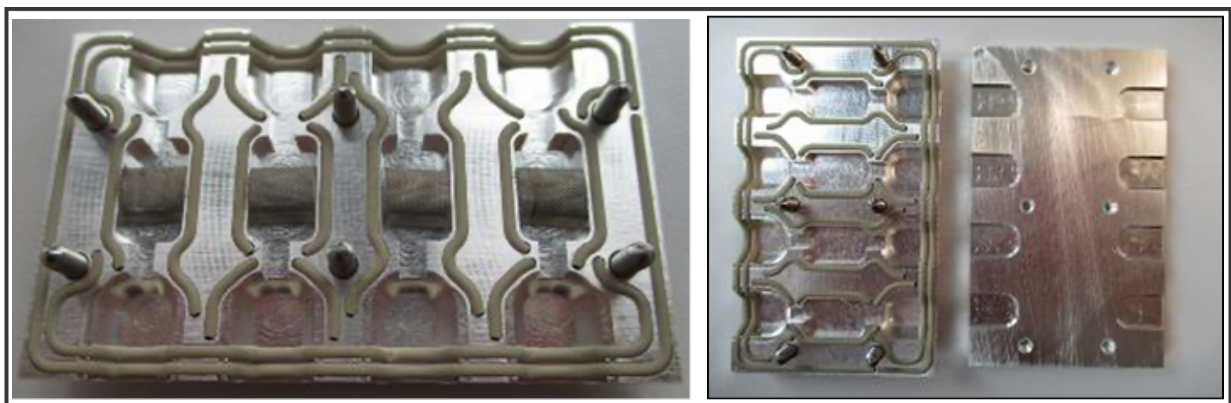


Figure 7: Silver-plated conducting cover used with the test board for 4x BFHKI filters.

The top cover contained a conductive sealing cord, and the device chambers were fitted with a compressible material to create a firm electrical contact with the top face of the BFHKL filter package. This so-called “EMI D-Profile” is built with a foam core covered with metallic woven fabric (2.0 x 6.4mm, LAIRD 4202-AE-221-07900), and must be mounted transverse to the channel. Otherwise, there may be a crosstalk path through the non-conductive foam core. 3x3mm pieces of damping material (0.76mm thick LAIRD Eccosorb GDS) were glued into the input and output channels to further suppress any crosstalk (Figure 8).

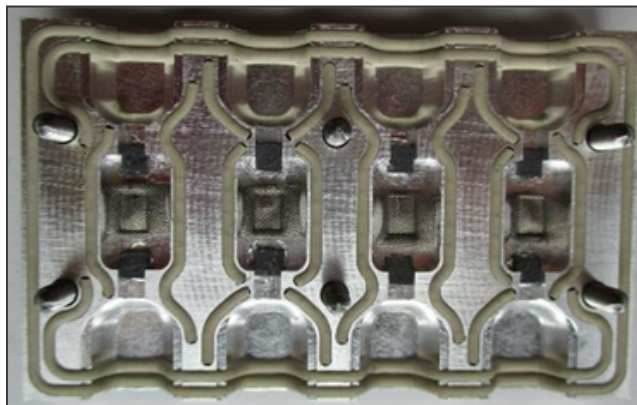


Figure 8: Cover pictured with compressible EMI D-Profile mounted in device chambers and damping material glued to input and output channels.

The strong imprints from the BFHKL units in Figure 8, indicate a good electrical contact to the top of the filter package, although the chamber depth may be a little low as dimensions were not optimized for these filters specifically.

Measurements

Multiple BFHKL models were measured in the test setup described above using a Rohde & Schwarz ZNA43 2-port vector network analyzer (Figure 9). Calibration was performed, and the DUT was de-embedded to the outer edge of the component footprint. The taper from the CPW line width to the pad's width is therefore still included in the measurement.



Figure 9: Measurement was performed using a Rohde & Schwarz ZNA43 2-port vector network analyzer

For each model of interest, S11 and S21 were swept over the full operating bandwidth under the following conditions:

- Open CPW test board without cover
- DUTs channelized with silver-plated conductive cover

- Channelized with cover and conductive die attach adhesive on outer edge of interposer

These measurements were compared to S-Parameters for the BFHK-series counterpart of each model as a baseline reference to study any deviations in each implementation of the BFHKI model on CPW. The S-Parameters were taken from the Mini-Circuits website and were measured on stripline test boards. The data presented below are compiled from the different measurements and superimposed on the same set of axes to easily compare the filter response under the different conditions tested.

Color Key:

- Red = Open
- Blue = With Cover
- Green = Cover + Adhesive
- Gray = BFHK on stripline (Mini-Circuits S-Parameters)

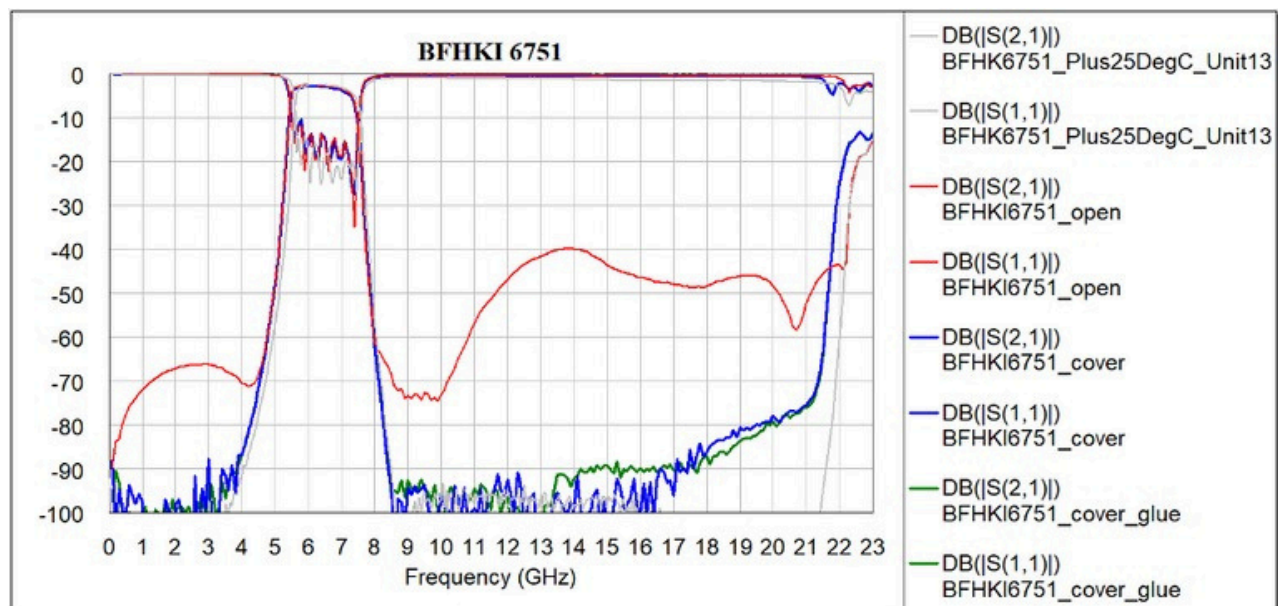


Figure 10: S11 and S21 plots for BFHKI-6751+ and BFHK-6751+ under various test conditions

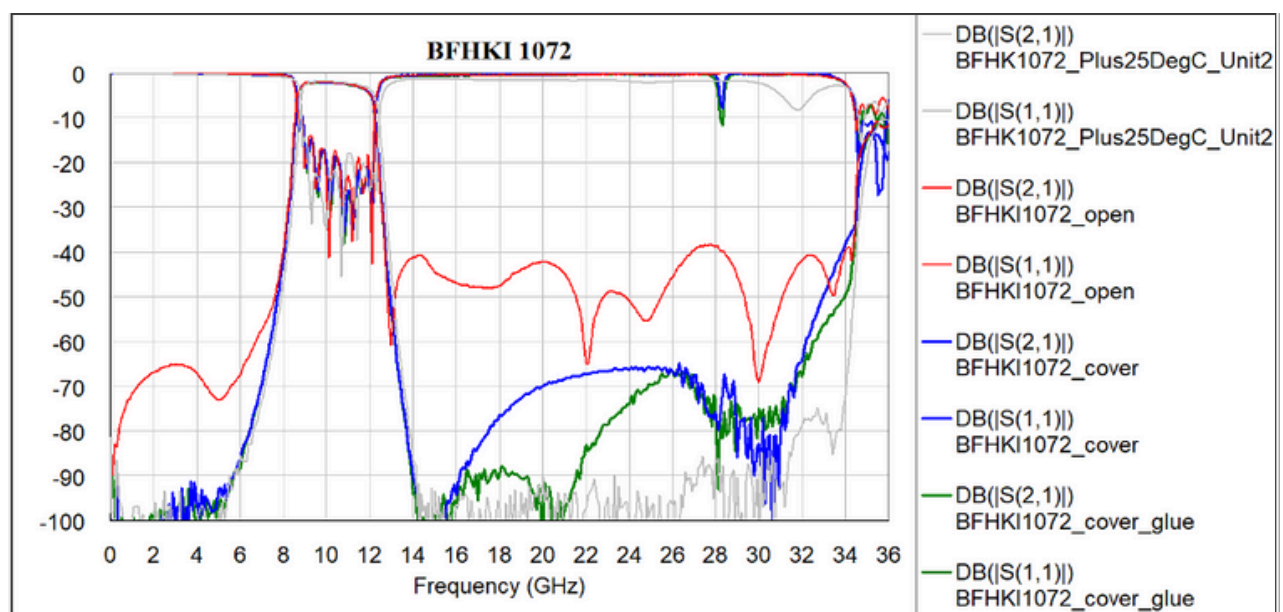


Figure 11: S11 and S21 plots for BFHKI-1072+ and BFHK-1072+ under various test conditions.

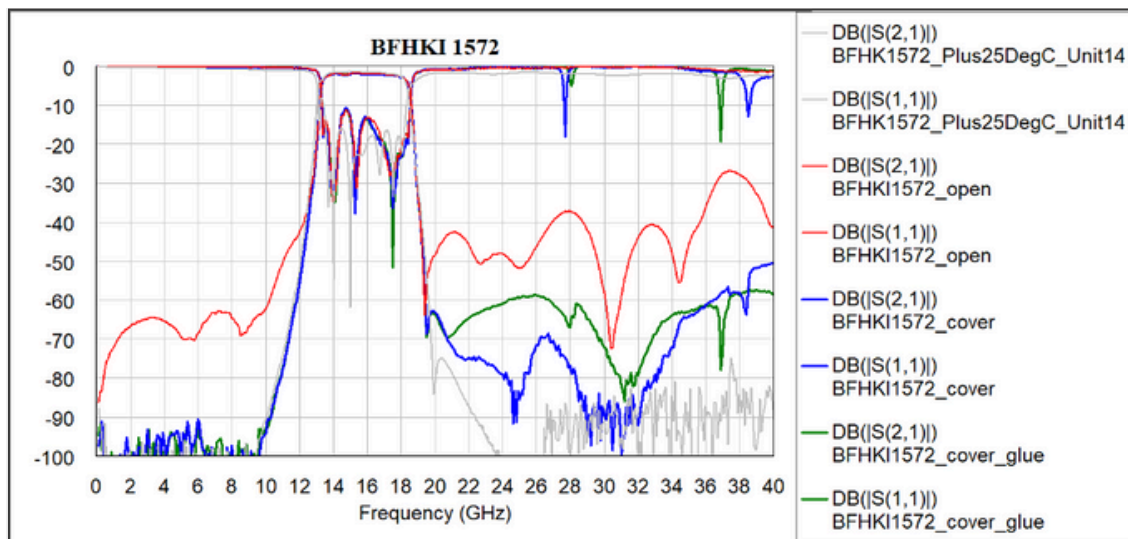


Figure 12: S11 and S21 plots for BFHKL1572+ and BFHK-1572+ under various test conditions.

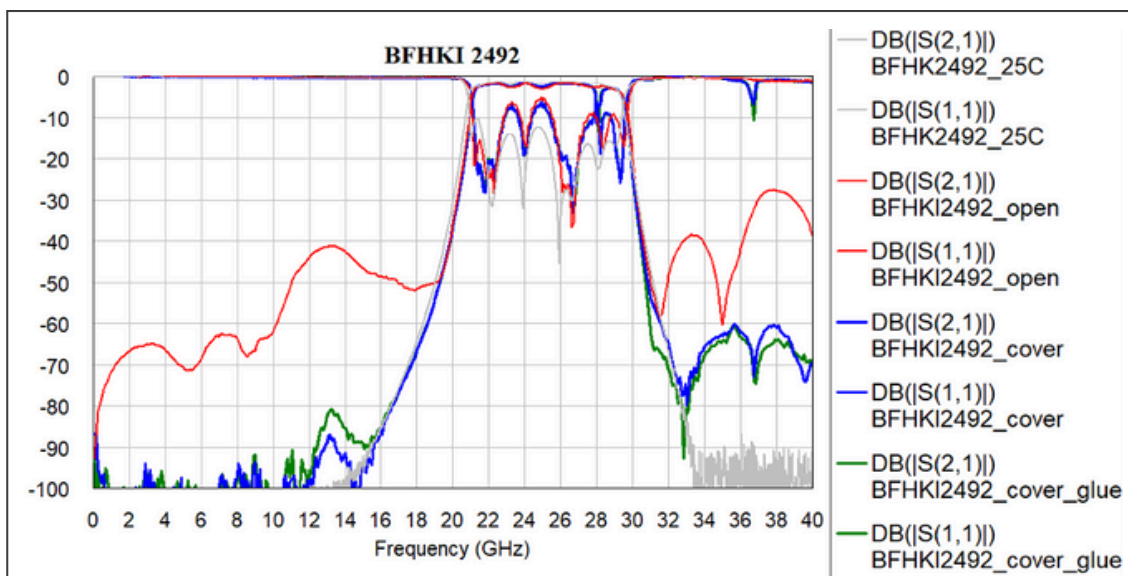


Figure 13: S11 and S21 plots for BFHKL2492+ and BFHK-2492+ under various test conditions. The dip in the passband at 28 GHz is caused by the cover.

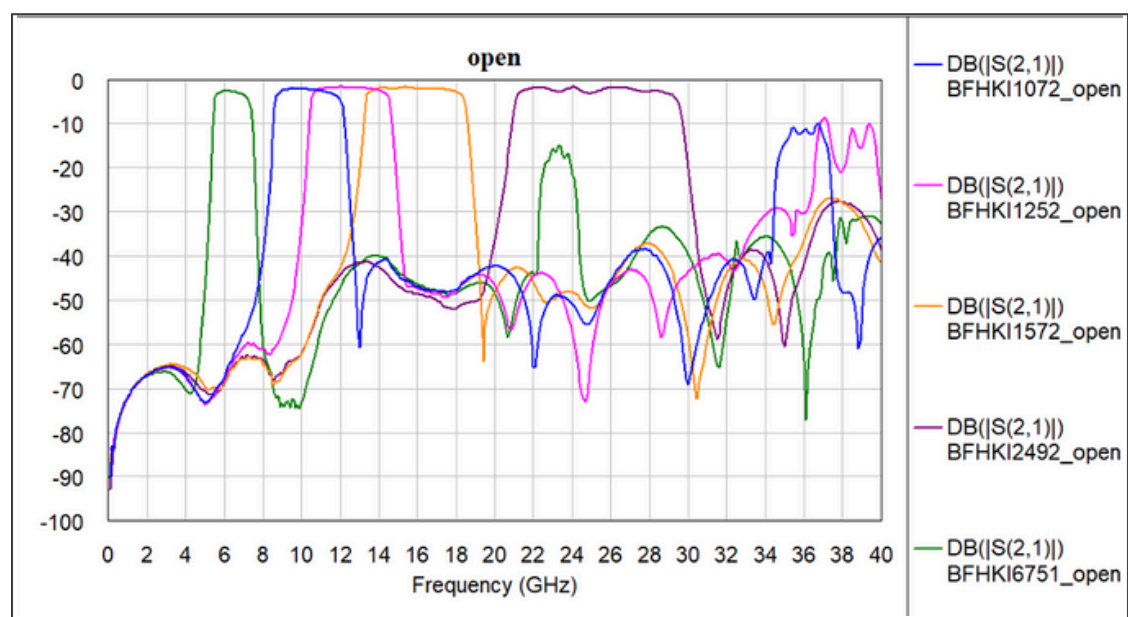


Figure 14: S21 plots for all BFHKL models tested on open CPW.

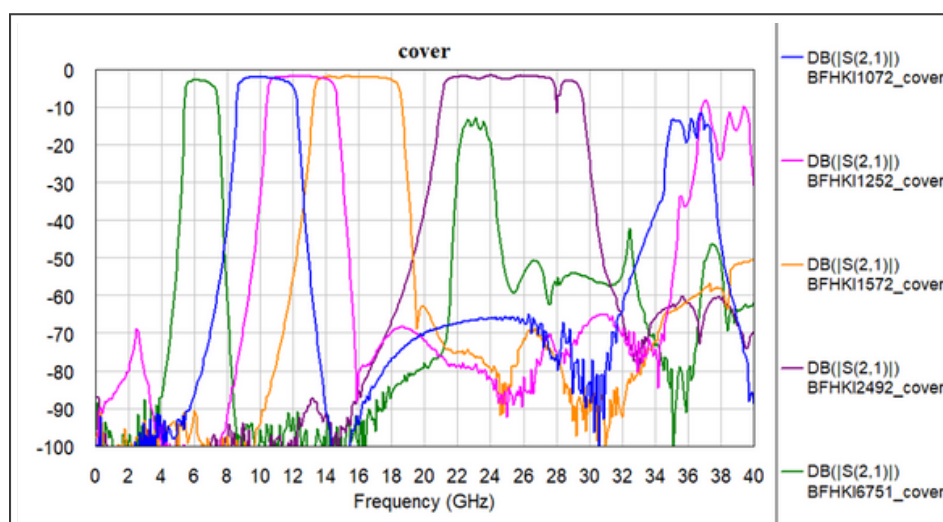


Figure 15: S21 plots for all BFHKI models tested with cover. Observe the dip at 28 GHz caused by the cover.

Discussion

In all cases, the addition of a channelized cover resulted in significant improvement in stopband rejection compared to the open CPW test case. Measurements of the BFHKI-series filters with the cover exhibited a response more closely approximating the reference case of the BFHK filter mounted directly on stripline with 90+ dB rejection floor in the lower stopband and upper stopband rejection ranging approximately from 60 to 90 dB depending on the model and the frequency of the upper stopband. More deviation was evident in the upper stopband, particularly at higher frequencies, but the channelized implementation still exhibited 60 to 70 dB rejection even at millimeter-wave range.

Application of conductive die attach adhesive achieved additional improvement in rejection in some cases, but caused resonances in others (e.g. BFHKI-1572+) resulting in poorer rejection. The cover may also cause resonances for higher passband frequencies, and a cover with chambers optimized to the dimensions of the BFHKI filter would presumably yield even better results.

Lastly, the measurements presented here exhibit poor return loss for passbands with higher center frequencies. A 10 mil test board rather than the 20 mil test board used here for filters with F_c higher than 18 GHz would likely correct this effect but was not included in this evaluation.

Conclusion

Mini-Circuits' BFHKI-series high-rejection LTCC filters with an interposer board were developed to extend the revolutionary capability of the BFHK series to CPW implementations with top-layer RF traces. While specifications on the datasheet exhibit degradation in rejection performance due to characterization on an open CPW test board, customers have demonstrated performance comparable to that of the BFHK filter on stripline PCB with channelization.

The channelization techniques presented in this paper are well-understood and widely used in the industry, making this a practical solution for customers using high-rejection LTCC filter technology in coplanar waveguide implementations.

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Filter Design Using Discrete Part-Value Optimization in Cadence AWR Design Environment

Optimization is often required to achieve desired performance when designing RF filters and other highfrequency circuits with today's simulation software tools. As an example, optimizing a lumped-element filter involves adjusting the values of its lumped components until the filter achieves an optimal frequency response. But once the component values have been determined via optimization, they may still need to be adjusted to the closest discrete, or "real-life," manufacturer part values. Depending on the design's complexity, this extra step can create a bit of extra legwork for the designer.

Of course, additional simulations must be performed after changing the optimized part values to the nearest available manufacturer part values. And if an optimized value falls roughly halfway between the two closest available manufacturer part values, one may need to experiment to determine which of the two values allows for better performance. In such cases, designers may need to carry out further refinements by adjusting interconnect dimensions to fine-tune the filter performance after setting the component values to the closest available manufacturer part values.

Fortunately, proper tools make it possible to perform discrete part-value optimizations in which component values are directly adjusted to optimal manufacturer part values. This optimization method eliminates the need for designers to manually adjust optimized component values to the closest available real-life part values, thereby cutting one step from the overall design process.

This application note explains how a discrete part-value optimization method can be leveraged to design a lumped-element bandpass filter designed using the Cadence AWR Design Environment (AWRDE). The design includes Modelithics measurement-based passive-component models, which enable simulations to accurately predict the filter's real performance. After completing an initial filter design, a discrete part-value optimization is carried out to achieve the desired frequency response. A yield analysis is also performed to determine the effect of component tolerances. At the conclusion, measured data is compared to the simulated results.

Beginning with the iFilter Synthesis Module

The AWRDE consists of several software tools used for developing RF/microwave products. Among them are Microwave Office for RF/microwave circuit design and the AXIEM 3D planar electromagnetic (EM) simulator. For this design example, both of these tools are used.

Various add-on modules can also be utilized within the AWRDE platform. One of them is iFilter, an integrated synthesis wizard used to develop RF/microwave filters. With iFilter, designers can synthesize lumped-element and distributed filters and then directly export them to Microwave Office for further analysis.

For this example, the design process begins with iFilter. Users open the iFilter module within AWRDE by double-clicking “iFilter Filter Synthesis,” which is located under “Wizards” in the “Project” browser. Starting a new filter design with iFilter prompts users to specify the “Passband” and “Realization.” Subsequently, “Main Filter Type” and “Options” must be specified (Fig. 1).

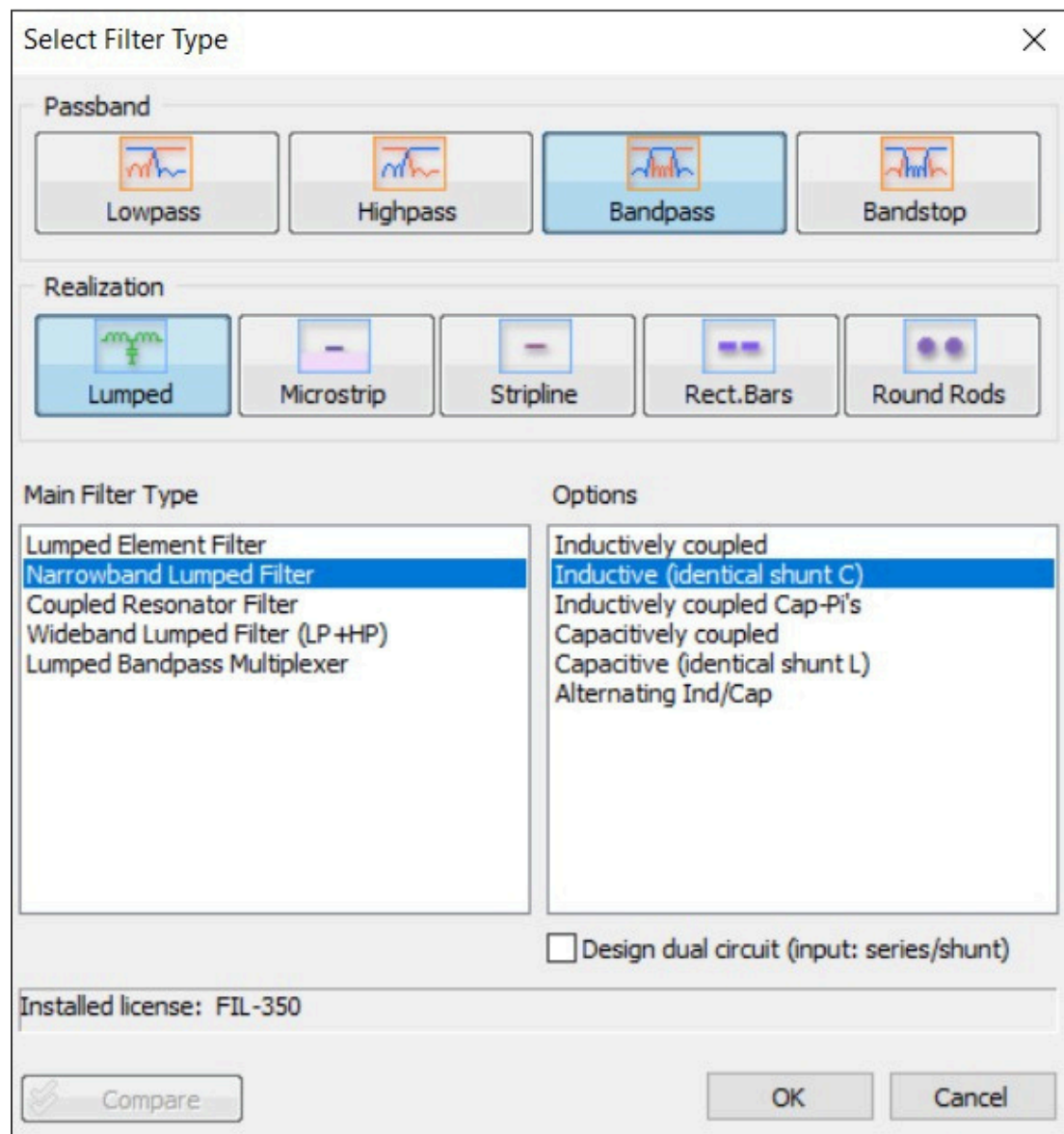


Figure 1. User interface when starting a design with iFilter.

For this example, “Passband” and “Realization” are specified as “Bandpass” and “Lumped,” respectively. Next, “Narrowband Lumped Filter” is chosen as the Main Filter Type. Subsequently, “Inductive (identical shunt C)” is chosen from the list of options.

Specifying the type of filter leads to the next user interface, which contains the filter schematic, frequency response, and several user-defined parameters (Fig. 2).

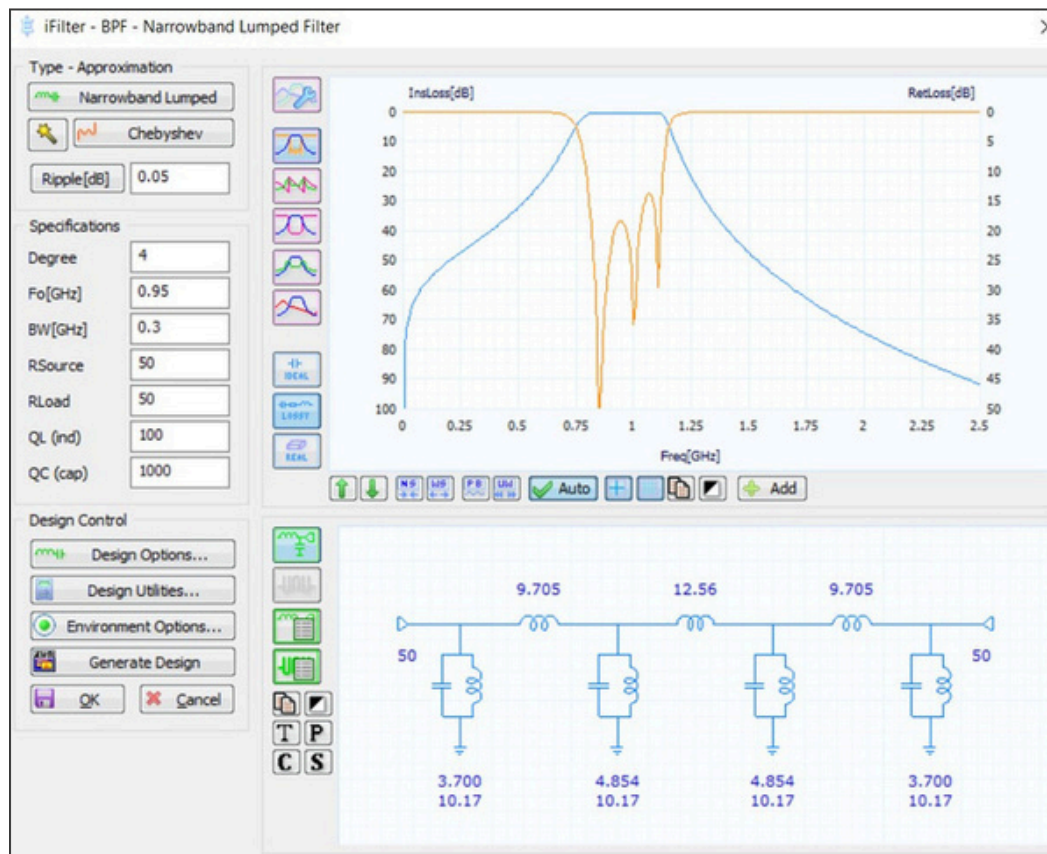


Figure 2. User-defined parameters, frequency response, and filter schematic.

For this design example, a Chebyshev filter with a center frequency of 950 MHz is specified. In addition, the bandwidth is set to 300 MHz.

Clicking “Generate Design” automatically creates a new schematic of the filter in Microwave Office (Fig. 3)

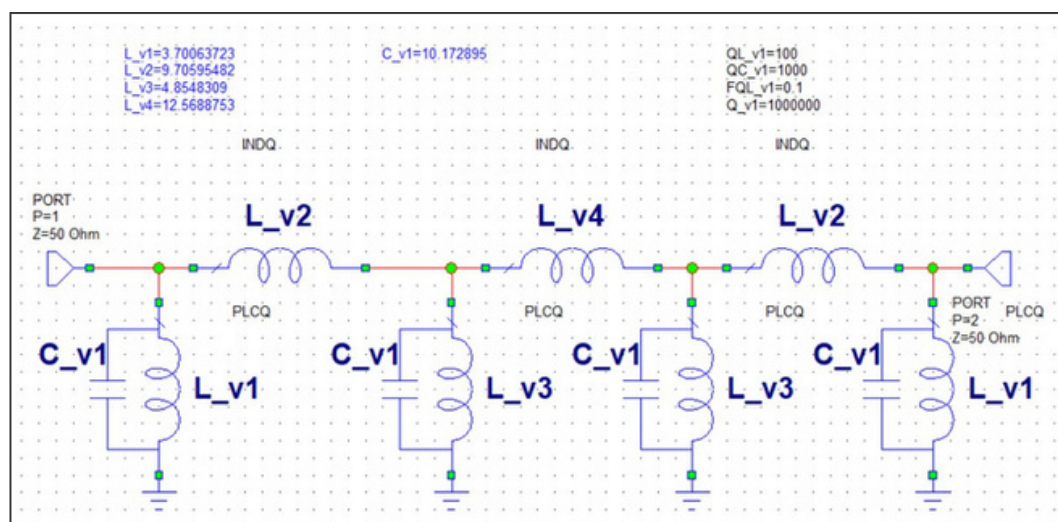


Figure 3. Generated Microwave Office schematic. Bold reference designations that correspond to the variables are shown for illustration.

Note that the tool automatically set several variables. The bold reference designations that correspond to the variables were added to the schematic for illustration. In this case, the variable C_v1 was created for all the capacitor values, while variables L_v1, L_v2, L_v3, and L_v4 were created for all inductor values. Table 1 lists these variables

and their corresponding values; Figure 4 shows the filter's simulated frequency response.

Table 1

L_v1	3.70063723 nH
L_v2	9.70595482 nH
L_v3	4.8548309 nH
L_v4	12.5688753 nH
C_v1	10.172895 pF

Variables generated by iFilter that represent the part values of the inductors and capacitors.

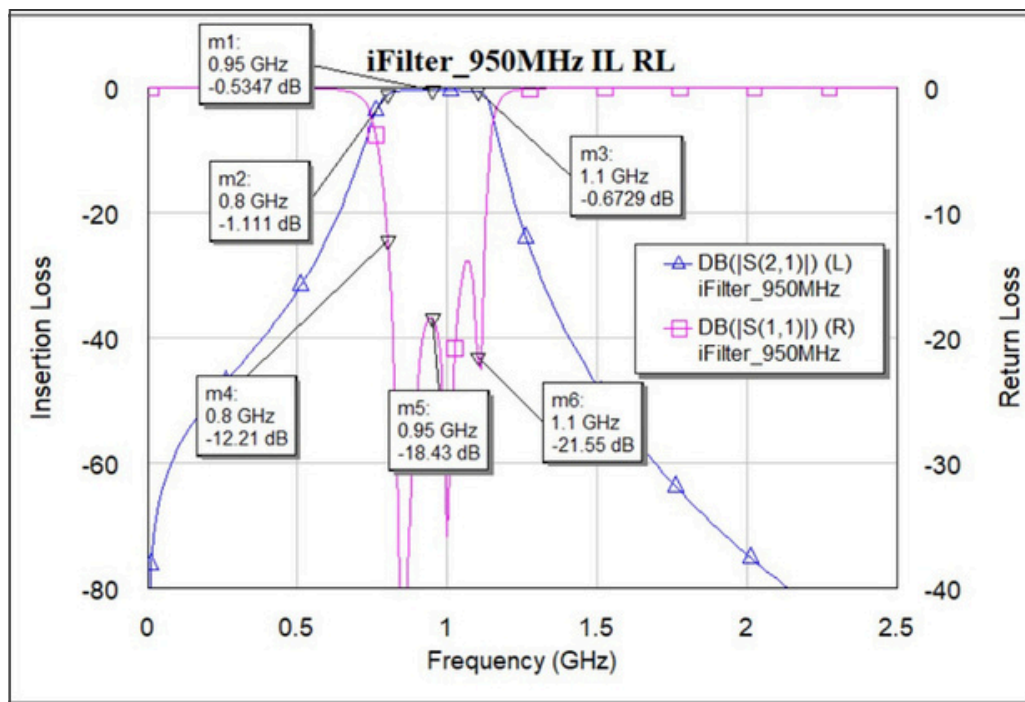


Figure 4. Simulated frequency response shows a center frequency of 950 MHz.

Incorporating Microstrip Transmission Lines and Modelithics Models

While the schematic shown in Figure 3 represents a good starting point for the design, the overall design process is far from complete. For one, to properly model a real filter, one must add microstrip interconnects to the schematic. Adding the microstrip interconnects makes it possible to generate a layout that can later be sent to a printed-circuit-board (PCB) manufacturer for fabrication.

One must also consider component modeling. In the schematic shown in Figure 3, the inductor and capacitor models are ideal closed-form elements with user-defined quality-factor (Q)

parameters. Alternatively, utilizing Modelithics measurement-based inductor and capacitor models makes it possible to predict the filter's performance more accurately. Modelithics models offer the benefit of both part-value and substrate scalability. These models accurately capture substrate-dependent parasitic behavior and offer advanced pad features for accurate EM/circuit co-simulations. Hence, Modelithics inductor and capacitor models will be used in place of the ideal elements seen in Figure 3.

Of course, one could simply modify the original schematic by adding the microstrip transmission lines and the Modelithics models. Another approach—and the one used in this case—is to duplicate the original schematic, allowing it to be left as a reference if needed. Figure 5 shows a new filter schematic created by duplicating the original one.

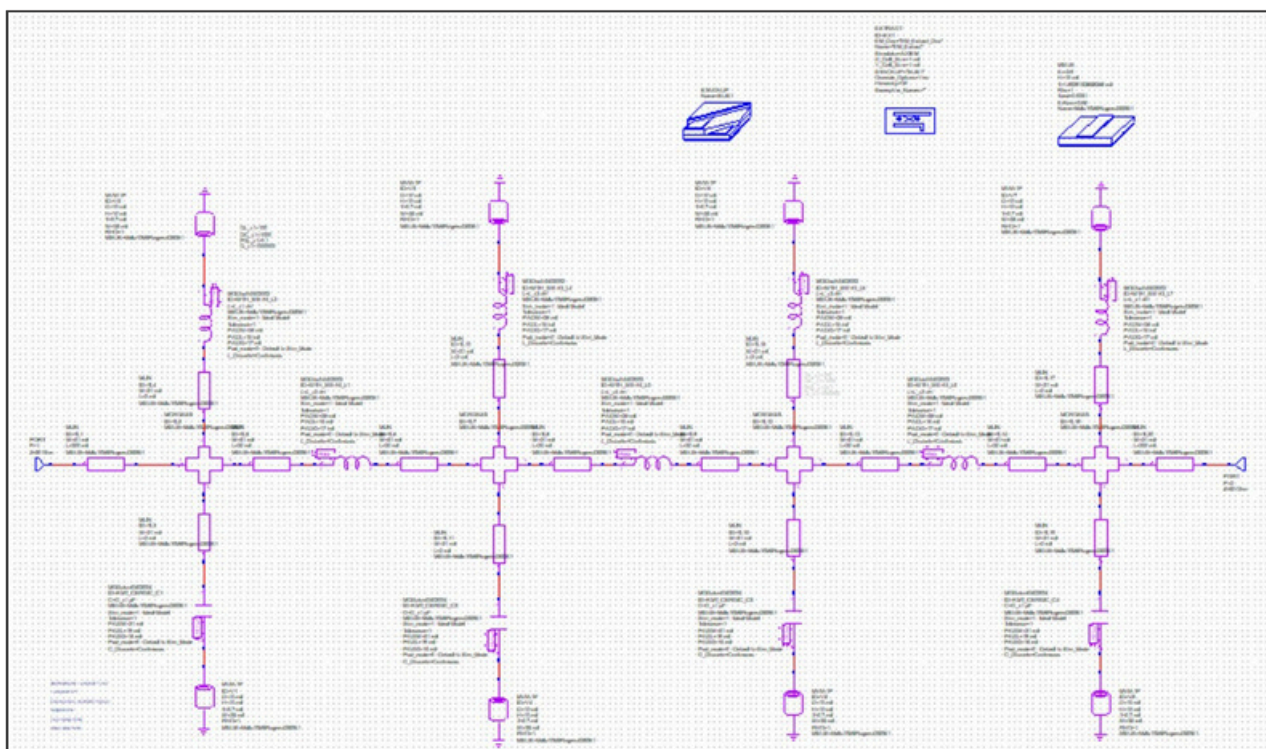


Figure 5. New filter schematic with microstrip interconnects and vias along with Modelithics passive-component models.

This new schematic contains all the necessary microstrip interconnects and vias. In addition, the ideal inductor and capacitor models are replaced with Modelithics models. For this example, the Würth Elektronik WE-KI 0402-size inductors are chosen for all the inductors. For the capacitors, Kemet's CBR04C 0402-size capacitors are used. The WE-KI 0402-size series covers an inductance range of 1 to 120 nH, while the CBR04C 0402-size series covers a capacitance range of 0.1 to 100 pF. Furthermore, the substrate used for this filter is 10-mil-thick Rogers RO4350B, which was implemented by placing the corresponding Modelithics substrate definition on the schematic.

The schematic shown in Figure 5 also includes an “EXTRACT” block and a “STACKUP” multi-layer substrate definition. These elements were added to the schematic via the “Create_Stackup” script, which can be accessed by selecting “Scripts” from the toolbar followed by “EM” and then “Create_Stackup.” These two elements together facilitate EM/circuit co-simulation by creating and configuring a metal geometry from the PCB layout of selected schematic elements. In this case, all the microstrip interconnects and vias are selected for extraction (Fig. 6).

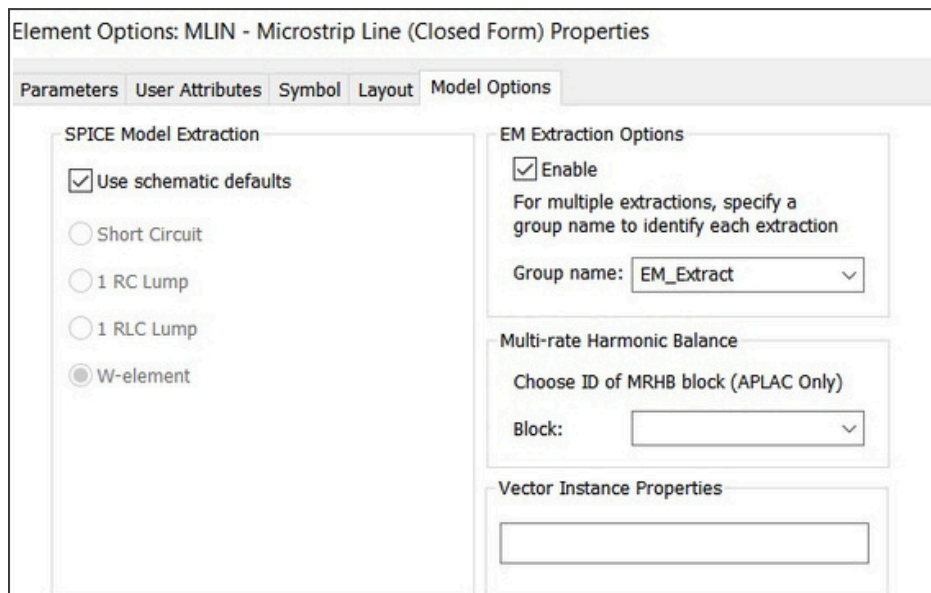


Figure 6. Microstrip element specified for EM extraction.

Therefore, these elements will be analyzed with the AXIEM EM simulator. The inductors and capacitors are the only elements from the schematic that are not extracted for EM analysis, because they will be analyzed within the Microwave Office circuit simulator. Figure 7 shows the extracted EM structure comprising of all the interconnects and vias.

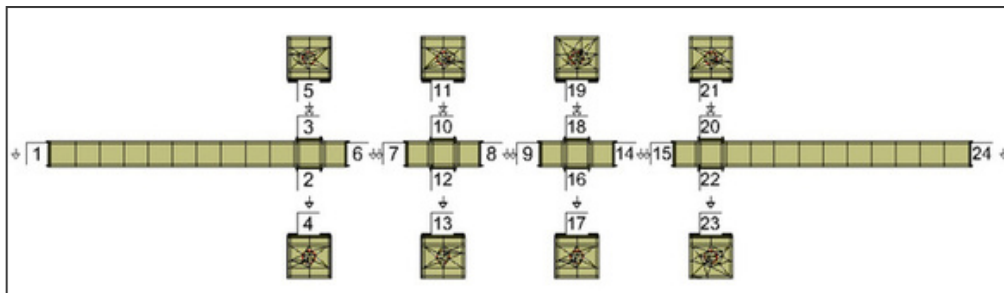


Figure 7. EM structure extracted using the "Create_Stackup" script.

Now that the EM extraction is properly configured, an EM/circuit co-simulation can be performed by simulating the schematic shown in Figure 5. As explained above, the microstrip interconnects and vias will be analyzed with the AXIEM EM simulator, while the components will be analyzed within the Microwave Office circuit simulator. Also, this schematic includes the same variables found in the original schematic (Table 1, again). The inductor and capacitor values of the Modelithics component models are set to the same variables as the corresponding components in the original schematic, meaning the component values are unchanged.

Some designers may want to see how the microstrip interconnects alone affect the filter's performance. To do so, set the "Sim_mode" parameter of all Modelithics models to 1. Setting a model's "Sim_mode" parameter to 1 enables it to simply behave as an ideal element, meaning that real-world parasitic, pad, and substrate effects are not considered. Thus, utilizing this setting for all models makes it possible to determine how the microstrip transmission lines alone affect performance, since the design is still employing ideal component models like the original schematic.

Figure 8 shows the simulated performance of the filter with the “Sim_mode” parameter set to 1 for all models. Notice how the response is shifted downward in frequency in comparison to the original frequency response (Fig. 4, again).

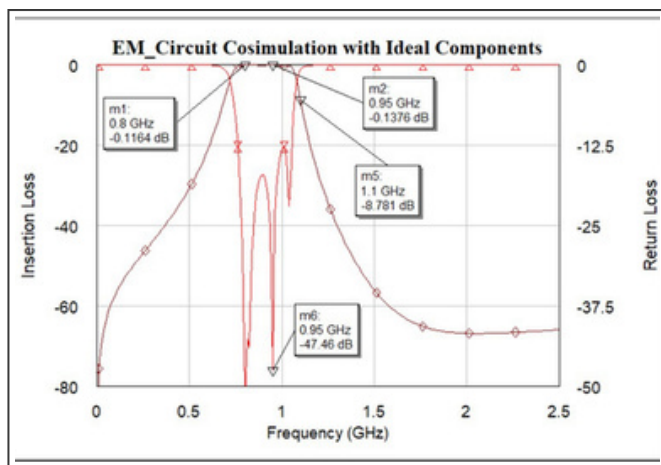


Figure 8. Simulated S21 and S11 with the “Sim_mode” parameter of all the Modelithics models set to 1. Adding the microstrip elements resulted in a downward shift in frequency in comparison to the original frequency response.

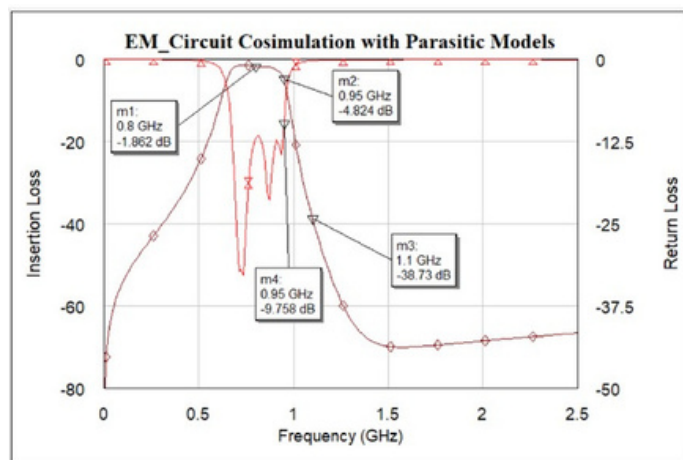


Figure 9. Simulation results with “Sim_mode” set to 0 for all Modelithics models. The filter shifts further downward in frequency and exhibits greater passband insertion loss.

The “Sim_mode” parameter of all models can now be set to 0. This setting enables a model to behave as a full parasitic model, thus accounting for all real-world parasitic, pad, and substrate effects. Figure 9 shows the frequency response after simulating the filter with “Sim_mode” set to 0 for all models.

It is evident that the response shifted further downward in frequency and is more lossy compared to the results shown in Figure 8. Specifically, the center frequency of the passband is approximately 800 MHz, a 150-MHz shift from the desired center frequency of 950 MHz. The filter also falls short of achieving the desired 300-MHz bandwidth.

Discrete Part-Value Optimization to the Rescue

Because the filter does not currently meet the design goals, the next step is to perform a discrete part-value optimization. This optimization technique will directly adjust the inductor and capacitor values to the optimal part values in the Würth Elektronik WE-KI and Kemet CBR04C part families, respectively.

To facilitate discrete part-value optimizations in the AWRDE, Modelithics offers .txt files for all passive component models. Each file contains a list of all manufacturer part values in the part family represented by the model. These files are located in the \Modelithics\Data folder, where is the installation folder (C:\Program Files (x86)\AWR\Foundry by default). The IND_WTH_0402_002.txt file contains all part values in the Würth Elektronik WE-KI family of inductors, while the CAP_KMT_0402_004.txt file contains the same information for the Kemet CBR04C capacitor family. Figure 10 displays the equations that must be added to the Microwave Office schematic (Fig. 5, again) to utilize these files for a discrete part-value optimization. The equations shown are based on the default file locations.

```

installDir="C:\Program Files (x86)\AWR\Foundry\Modelithics"
dataDir=installDir + "\Data\"

L_vector=vfile(dataDir + "IND_WTH_0402_002.txt")
C_vector=vfile(dataDir + "CAP_KMT_0402_004.txt")

L_v1=L_vector[8]=3.6
L_v2=L_vector[20]=9.5
L_v3=L_vector[11]=4.7
L_v4=L_vector[24]=13

C_v1=C_vector[100]=10

```

Figure 10. Equations added to the Microwave Office schematic to enable a discrete part-value optimization.

Note that the IND_WTH_0402_002.txt and CAP_KMT_0402_004.txt files generate two vectors, “L_vector” and “C_vector,” respectively. As a result, the “L_vector” element contains each of the part values in the Würth Elektronik WE-KI part family, while the “C_vector” element contains each of the part values in the Kemet CBR04C part family. To explain further, the Würth Elektronik WE-KI model encompasses a total of 48 inductance values that range from 1 to 120 nH. Thus, the first vector index in the “L_vector” element corresponds to 1 nH, while the 48th and final vector index corresponds to 120 nH. The Kemet CBR04C model covers a capacitance range of 0.1 to 100 pF, with a total of 124 capacitance values. Therefore, the first vector index in the “C_vector” element corresponds to 0.1 pF, while the 124th and final vector index corresponds to 100 pF.

From the values shown in the bottom portion of Figure 10, one can see that the variables are no longer equal to the initial values of Table 1. Rather, they are now set to the vector indices that correspond to the manufacturer part values closest to the initial values. Users may easily determine these indices by copying the part values from the .txt files and pasting them into a spreadsheet (Fig 11).

[↑ View Table of Contents ↑](#)

	A	B	C	D	E
1	1			0.1	
2	1.9			0.2	
3	2			0.3	
4	2.2			0.4	
5	2.4			0.5	
6	2.7			0.6	
7	3.3			0.7	
8	3.6			0.8	
9	3.9			0.9	
10	4.3			1	
11	4.7			1.1	
12	5.1			1.2	
13	5.6			1.3	
14	6.2			1.4	
15	6.8			1.5	
16	7.5			1.6	
17	8.2			1.7	
18	8.7			1.8	
19	9			1.9	
20	9.5			2	
21	10			2.1	
22	11			2.2	
23	12			2.3	
24	13			2.4	
25	15			2.5	
26	16			2.6	
27	18			2.7	
28	19			2.8	
29	20			2.9	

Figure 11. The part values from the .txt files can be pasted into a spreadsheet (columns A and D) to determine the vector indices.

Simulating the filter produces the frequency response shown in Figure 12. The results are similar—but not quite identical—to the simulated results of the filter with the initial part values (Fig. 9, again). Table 2 lists the manufacturer part values used for this initial simulation.

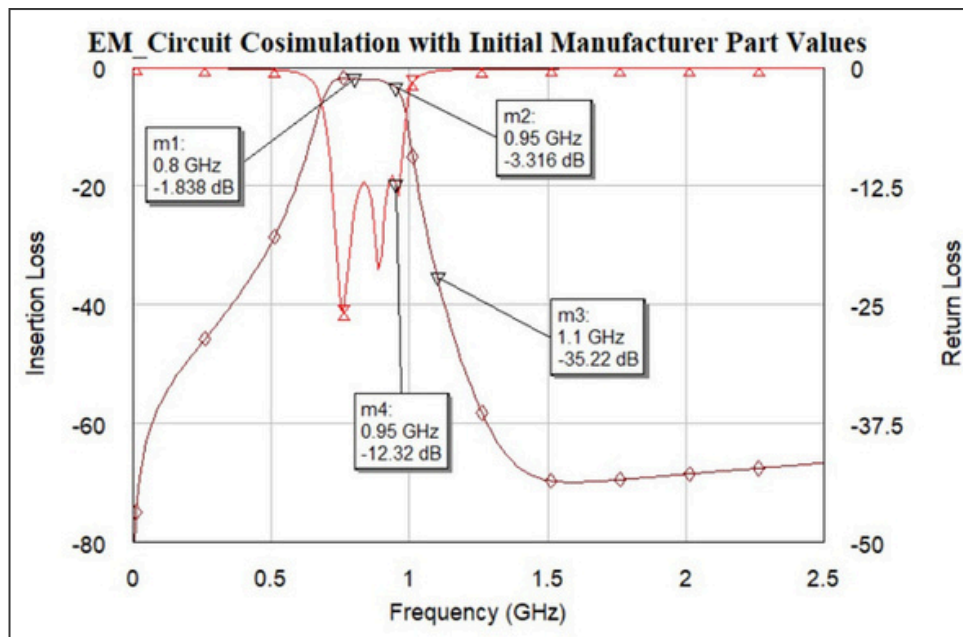


Figure 12. Simulated results after changing the filter's initial part values to the closest manufacturer part values

Table 2

L_v1	3.6 nH
L_v2	9.5 nH
L_v3	4.7 nH
L_v4	13 nH
C_v1	10 pF

Manufacturer part values closest to the initial values.

It is now time to perform a discrete part-value optimization. To do so, the variables L_v1, L_v2, L_v3, L_v4, and C_v1 must be specified for optimization in the “Variable Browser” (Fig. 13)

Document	Element	ID	Parameter	Value	✓ Tune	✓ Optimize	✓ Constrain	Lower	Upper	Step Size	Tag
MDLX F											
MDLX Filter EM Extraction	EQN		L_v1	L_vector[8]	☐	☒	☐				
MDLX Filter EM Extraction	EQN		L_v2	L_vector[20]	☐	☒	☐				
MDLX Filter EM Extraction	EQN		L_v3	L_vector[11]	☐	☒	☐				
MDLX Filter EM Extraction	EQN		L_v4	L_vector[24]	☐	☒	☐				
MDLX Filter EM Extraction	EQN		C_v1	C_vector[100]	☐	☒	☐				
MDLX Filter EM Extraction	MODIwth0402002	WTH_WE-KI_L3	Tolerance	1	☐	☐	☐				
MDLX Filter EM Extraction	MODIwth0402002	WTH_WE-KI_L3	PADW	26	☐	☐	☐				
MDLX Filter EM Extraction	MODIwth0402002	WTH_WE-KI_L3	PADL	15	☐	☐	☐				

Figure 13. To execute a discrete part-value optimization, the variables that represent the part values of the inductors and capacitors must be specified for optimization in the “Variable Browser.”

Next, the optimization goals must be set. Over the passband of 800 to 1,100 MHz, the optimization goals are an S21 value greater than -2 dB and an S11 value less than -12 dB. For the lower and upper rejection bands, the goal is an S21 value of less than -20

dB. Finally, upon opening the optimization user interface, “Discrete Local Search” must be selected from the list of optimization methods.

Figure 14 shows the filter’s simulated frequency response after performing the discrete part-value optimization, while Table 3 lists the optimized manufacturer part values.

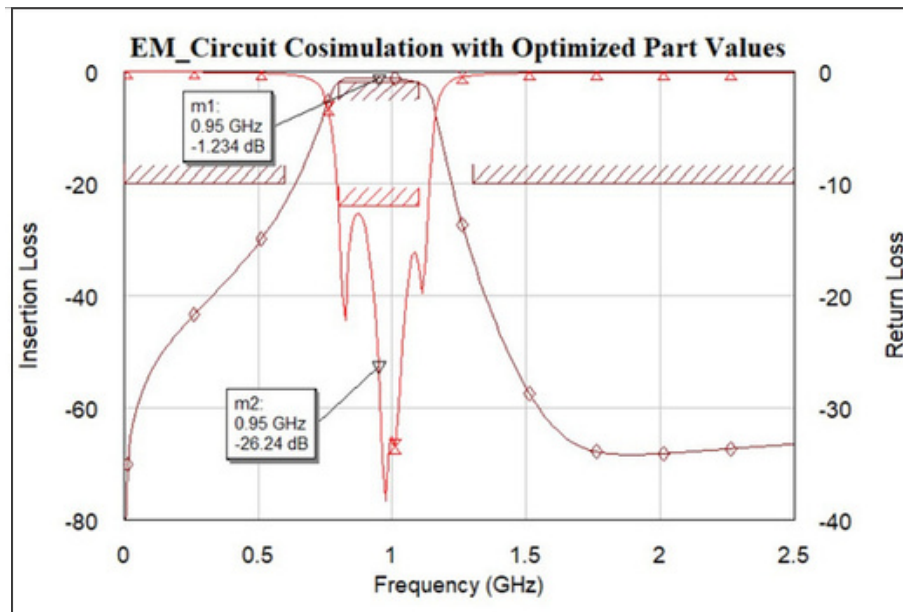


Figure 14. Post-optimization frequency response. The response reveals a center frequency of 950 MHz along with the desired bandwidth.

Table 3

L_v1	3.6 nH
L_v2	9 nH
L_v3	5.6 nH
L_v4	11 nH
C_v1	7.0 pF

Final optimized manufacturer part values.

The filter now achieves the desired performance. Keep in mind that one may fine-tune interconnect dimensions to tweak a design so as to achieve the desired performance. In this case, that step wasn’t necessary.

A yield analysis is the final step in the design process. This analysis is performed to determine the sensitivity of the design with respect to part-value changes due to component tolerances. The tolerances of the components in the optimized design can be obtained from the Würth Elektronik and Kemet websites. L_v1 (3.6 nH) is specified for a tolerance of $\pm 5\%$, while the tolerances of L_v2 (9 nH) and L_v3 (5.6 nH) are each specified for ± 0.2 nH. L_v4 (11 nH) is specified for a tolerance of $\pm 2\%$. Finally, C_v1 (7.0 pF) is specified for a tolerance of ± 0.1 pF.

The Modelithics models include a “Tolerance” parameter that can be utilized for yield simulations. To perform a yield analysis, users must check the “Use Statistics” checkbox next to the “Tolerance” parameter and then specify the tolerance in percentage as well as the distribution (Fig. 15).

Element Options: MODIwth0402002 - IND_WTH_0402_002 Wurth WE-KI 0402 substrate scalable Global inductor model Properties (Showing)

Parameters User Attributes Symbol Layout Model Options												
Name	Value	Unit	Tune	Optimize	Constrain	Lower	Upper	Step Size	Use Statistics	Yield Optimize	Tolerance	Distribution
ID	WTH_WE-KI_L3											
L	L_v4	nH										Uniform
MSUB	Mdx10MilRogers4350B1											
Sim_mode	0 - Full Parasitic Model											
Tolerance	1		☐	☐	☐				☒	☐	2%	Normal
PADW	26	mil	☐	☐	☐				☐	☐		
PADL	15	mil	☐	☐	☐				☐	☐		
PADG	17	mil	☐	☐	☐				☐	☐		
Pad_mode	0 - Default to Sim_Mode											
L_Discrete	Continuous		☐	☐	☐				☐	☐		

Figure 15. Model parameters with the “Tolerance” parameter activated.

In this case, a normal (Gaussian) distribution was specified for all components. Figure 16 shows the results of the yield analysis.

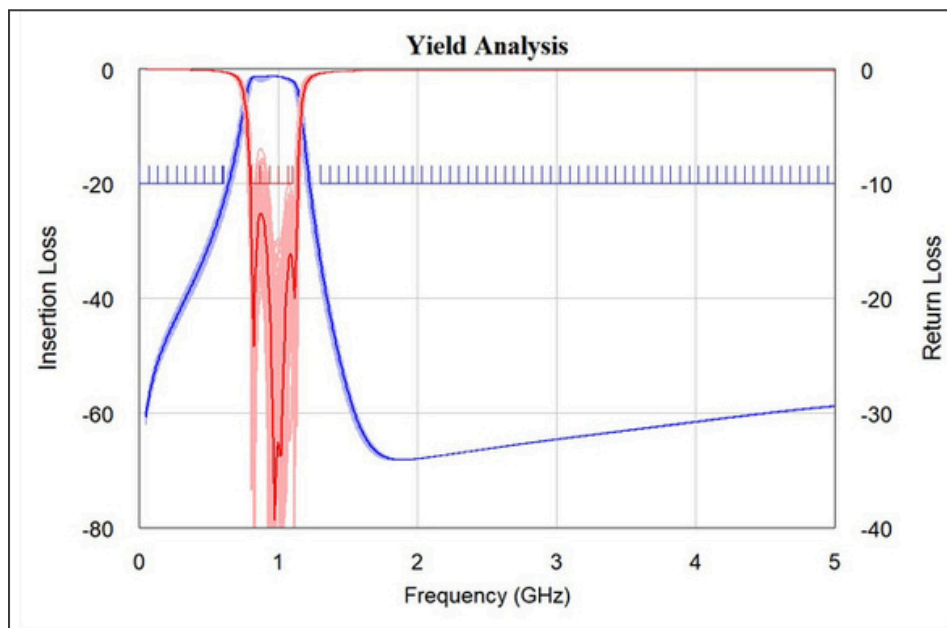


Figure 16. Yield analysis simulation results showing S21 and S11.

It is clear that part-value tolerances have a greater effect on the passband, as the results show a noticeable variation in S11 and a slight variation in S21. In contrast, part-value tolerances have little effect on out-of-band performance.

Measured Data Versus Simulated Results and Closing

The final step is to validate the design by building and measuring the filter. Two filters were assembled using the same Würth Elektronik WE-KI inductors and Kemet

CBR04C capacitors from the final simulation (Fig. 17). Figure 18 shows the measured data of both filters.

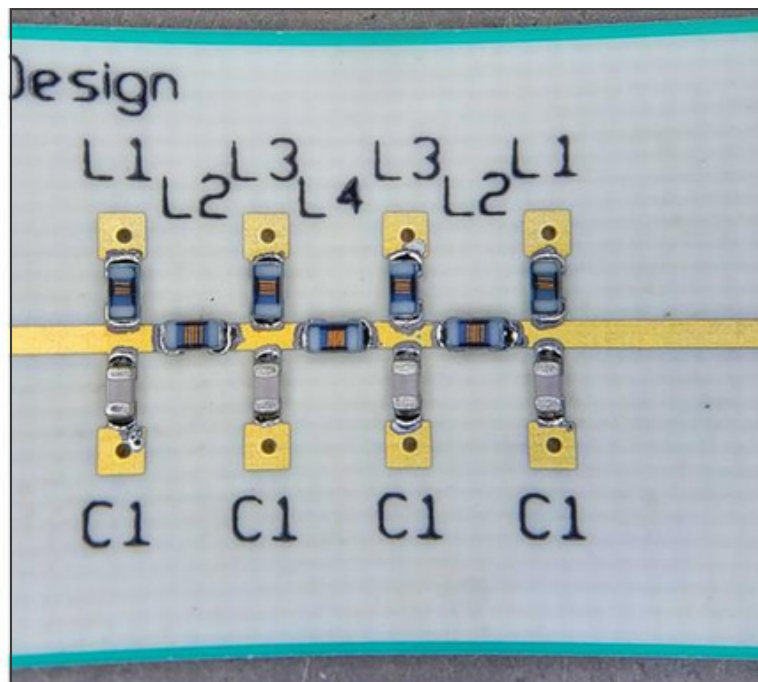


Figure 17. One of two filters built.

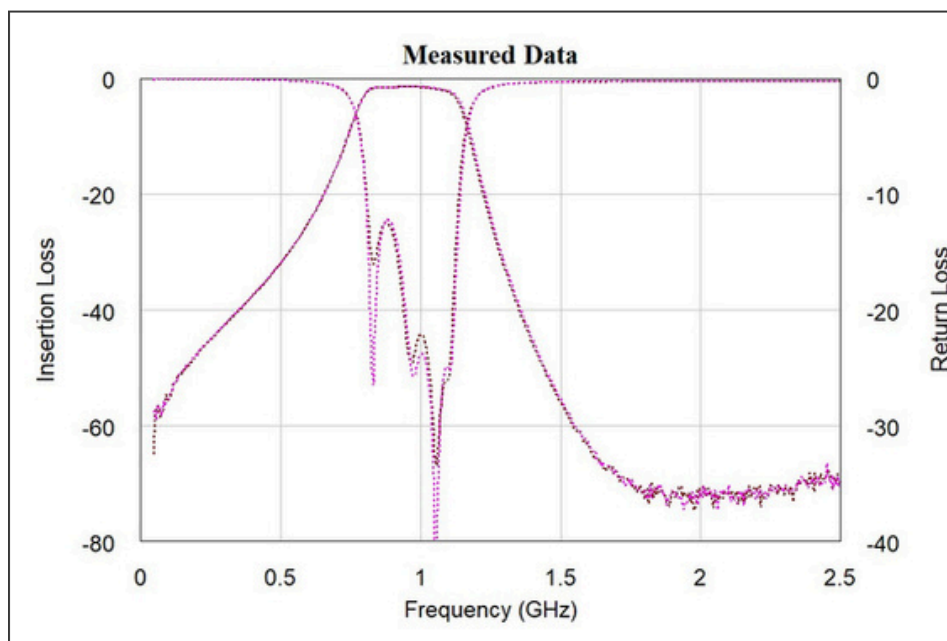


Figure 18. Measured S21 and S11 of both filters.

As a comparative analysis, the “Sim_mode” parameter of all the Modelithics models can be set to 1 to see how the performance changes when using ideal models with the final optimized part values. Figure 19 shows the simulated S21 with every model still set to the optimized part value but with the “Sim_mode” parameter of each now set to 1. The measured data of both filters is also shown. The simulated results clearly do not agree with the measured data, as the simulated frequency response is shifted higher in frequency. The simulated bandwidth is also wider.

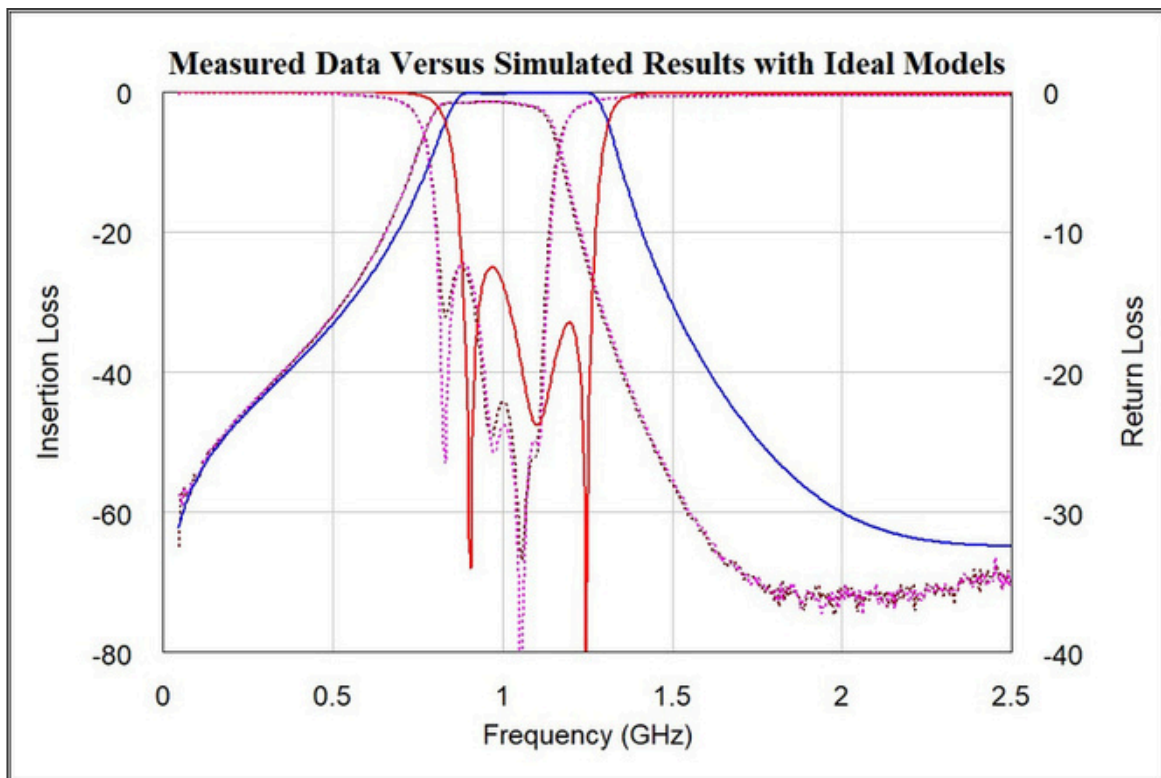


Figure 19. Measured data (dashed traces) and simulation results (solid lines) when models are set to ideal mode with the optimized part values.

Figures 20 and 21 show the measured data along with the final simulated results when the “Sim_mode” parameter is set to 0 for all models. The measured data corresponds to the simulated results, thereby illustrating the benefit of using real-world parasitic models. The design process is therefore validated.

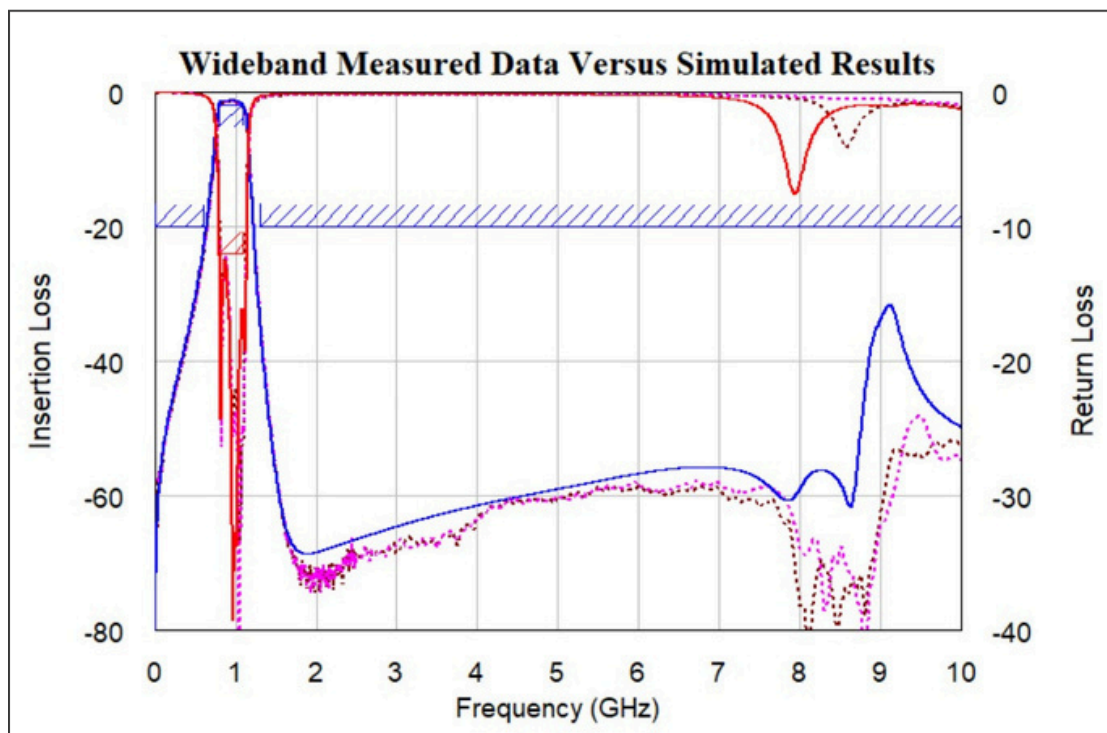


Figure 20. Wideband measured data for S21 and S11 (dashed traces) along with the final simulated results (solid traces) when the “Sim_mode” parameter of all models is set to 0.

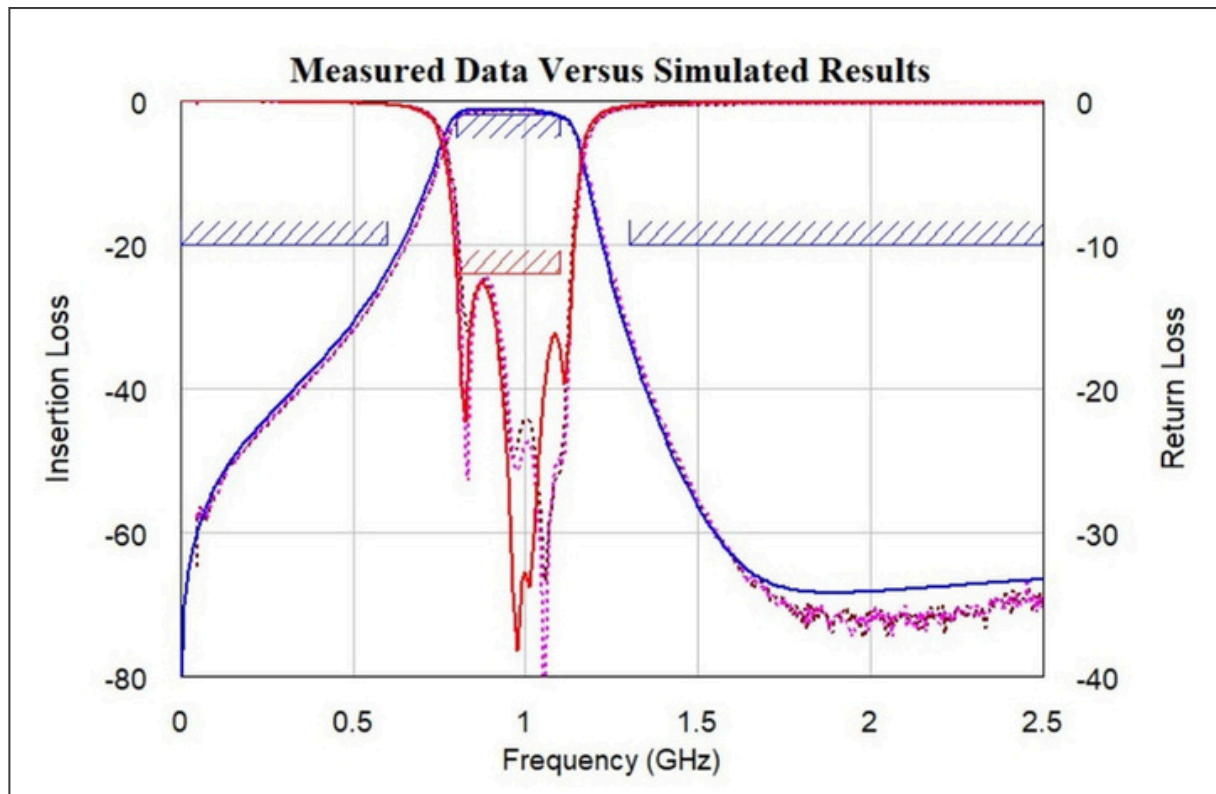


Figure 21. Narrowband measured data for S21 and S11 (dashed traces) along with the final simulated results (solid traces) when the "Sim_mode" parameter of all models is set to 0.

In closing, utilizing Modelithics measurement-based models combined with a discrete part-value optimization method within the Cadence AWR Design Environment can be an effective way to design filters. The approach pinpoints exactly which manufacturer part values are best suited for a design. A yield analysis can also be performed to determine the sensitivity with respect to component tolerances.

Advanced BAW Filter Technology and Its Impact on 5G

Abstract

The global rollout of 5G, and the rapid expansion of the Internet of Things, create significant new RF filtering challenges. Qorvo's bulk acoustic wave (BAW) filter technology is advancing to overcome these challenges. Among the chief developments: BAW filters are evolving to support higher frequencies and increased bandwidth for the new 5G and Wi-Fi band expansion. Complex multi-filter modules (multiplexers and antennaplexers) are being used to address RF system challenges, particularly with 5G. Additionally, a smaller uBAW (micro-BAW) form factor helps squeeze complex RF front end (RFFE) architectures into the limited space available in handsets and Internet of Things (IoT) devices. At the same time, Qorvo's BAW technology helps to mitigate the problem of heat dissipation associated with higher frequencies and smaller form factors.

Introduction

Qorvo's BAW RF filters play indispensable roles in wireless devices and infrastructure. They ensure isolation between the ever-growing number of frequency bands and offer low insertion loss to support device performance requirements. However, the global rollout of 5G, and the evolution of other wireless standards such as Wi-Fi, create new filtering challenges that are beyond the capabilities of traditional acoustic filters. Significant advances in BAW technology were required to solve these new challenges.



Higher Frequencies. 5G and other RF standards including Wi-Fi and enhanced Licensed Assisted Access (eLAA) are using frequencies up to 7 GHz, well above the range traditionally supported by acoustic filters.



Increased Bandwidth. 5G is all about increased bandwidth to achieve dramatically higher data speeds and network capacity. BAW filters must therefore support the increased bandwidth allocated to 5G bands.



Lower Insertion Loss. Demand for filters with lower insertion loss has increased since the introduction of 5G. Lower loss filtering means better performance, which is needed to attain the 5G wireless system specifications.



Increased Rejection. With the additional 5G spectrum re-farming and crowding, an increase for filter technology to help coexistence is a must.



Heat Dissipation and Reliability. With the onset of higher frequency ranges comes the challenge of dissipating heat. BAW filters are the best small-size technology to dissipate heat, thereby helping to increase reliability of the filter and the RF system.



Reduced Size. As RFFE complexity continues to increase rapidly, BAW filters must become smaller and more integrated to fit into the limited space available in devices.



Complex Filtering Requirements. The increasing use of multi-filter modules, such as multiplexers and antennaplexers to address carrier aggregation (CA) requirements, creates new filter performance challenges.



Accurate Simulation. Unprecedented accuracy in simulation and modeling is needed to ensure that filters can be produced to the required specifications within short timescales.

Key Properties of BAW

The basic structure of BAW resonators consists of a thin-film layer of piezoelectric material sandwiched between two metal thin-film electrodes. The voltage or electrical field between the two electrodes excites an acoustic wave (see Figure 1). The wave bounces between the surfaces of the two electrodes, forming an acoustic cavity

between the top surface of the upper electrode and the bottom surface of the lower electrode. The resonant frequency is inversely proportional to film thickness and material properties; it is determined both by the thickness of the piezolayer and by the thickness and material properties of the electrodes.

In BAW solidly mounted resonators (SMR), alternating solid layers of low and high acoustic impedance below the piezoelectric and electrode layers act as an acoustic Bragg reflector to confine energy in the vertical direction. The Bragg layers provide reflectance that approaches the performance of a free surface. The perimeter of the resonator is engineered to confine acoustic energy laterally, minimize acoustic leakage and suppress unwanted modes.

The RF properties of a BAW filter are directly related to the thickness and material properties of the piezoelectric, dielectric and metal films. Precise thicknesses are achieved by optimized thin-film deposition and ion milling based on metrology and electrical measurements. A higher resonant frequency can be achieved by removing small amounts of the top layer film using ion milling.

The most common piezoelectric material used for BAW is Aluminum Nitride (AlN). AlN provides the best balance of performance, reliability and manufacturability. The AlN piezoelectric layer must have a high level of crystal orientation to achieve optimum electromechanical coupling.

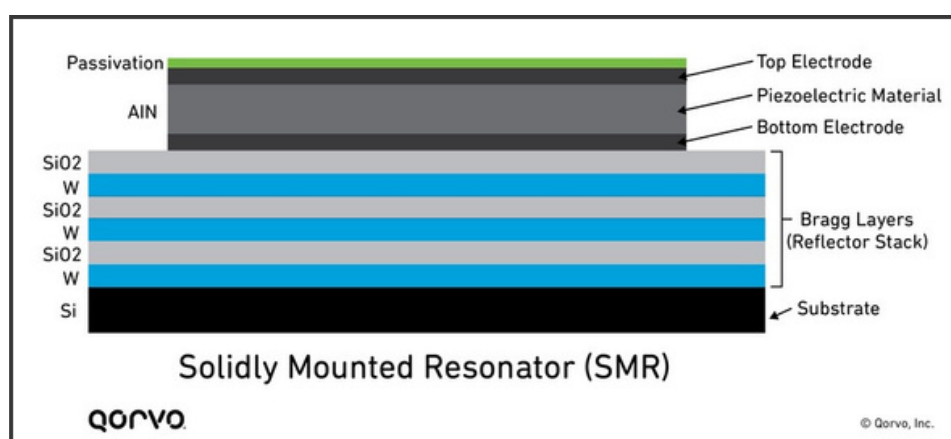


Figure 1. Cross section of BAW SMR filter.

A filter's insertion loss is defined by several factors, including the losses in its individual resonators. There are several possible loss mechanisms, but the most frequently discussed is the loss of energy due to acoustic leakage, as shown in Figure 2. However, most of the losses come from viscous losses in the materials.

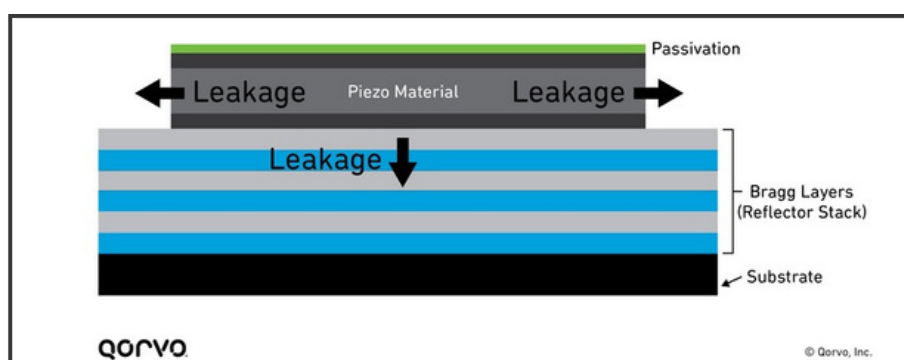


Figure 2. Possible acoustic leakage mechanisms in SMR BAW.

The Importance of Electro-Mechanical Coupling

Superior coupling is imperative to meet high-frequency passband requirements for 5G and higher frequency Wi-Fi. The achievable width of the filter passband is determined by the electromechanical coupling coefficient, k_{eff}^2 . Maximizing k_{eff}^2 is one of the biggest challenges in thin-film BAW design.

The k_{eff}^2 of a BAW resonator can be obtained from the resonance and anti-resonance frequencies measured from its impedance response. The design of the resonator layers can greatly enhance or degrade the coupling.

Filter Topologies

Networks of resonators can be designed to achieve various filter characteristics. BAW filters can be classified into two major topology categories: ladder and lattice. A lattice is a balanced topology, while a ladder is an unbalanced topology (see Figure 3).

Each topology has its advantages and disadvantages; however, the ladder topology is the most common. To achieve the required bandwidth, the shunt elements are tuned to a lower frequency than the series elements. The out-of-band rejection is determined by the number of elements and the net capacitor divider. To meet a desired rejection there is typically an optimum topology.

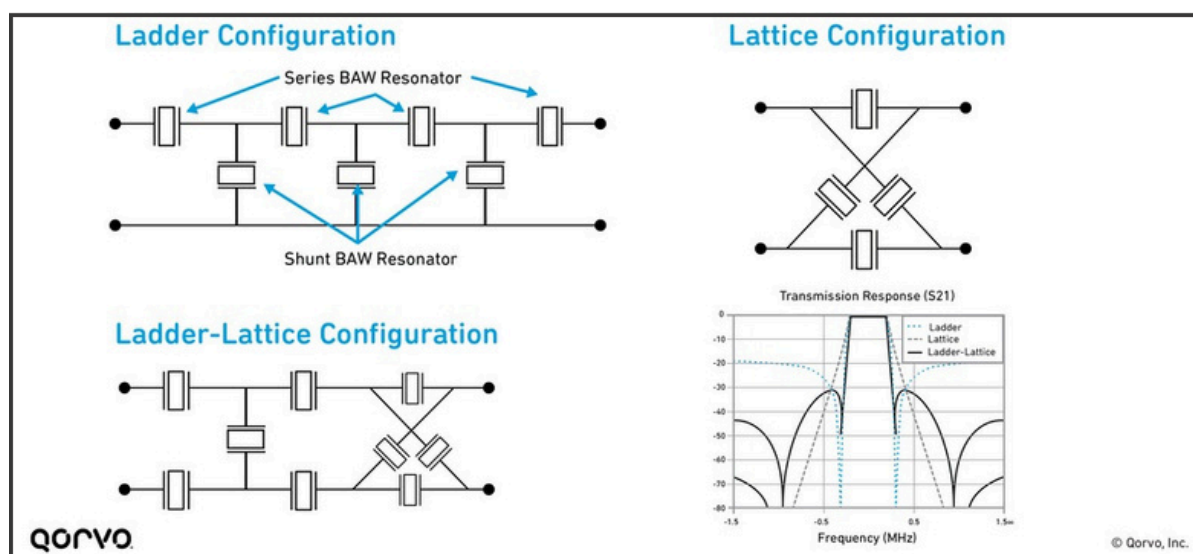


Figure 3. Ladder, lattice and ladder-lattice configurations.

High-Frequency BAW

5G uses new frequency bands that are well above those traditionally used for 4G LTE, and above the range customarily supported by BAW filters. These include new 5G Frequency Range 1 (FR1) and Frequency Range 2 (FR2) bands as shown in Figure 4 below. The upper limit of the Wi-Fi spectrum is also being extended above 7 GHz. Evolving BAW technology to support these considerably higher frequencies requires solving several technology challenge. Such as:

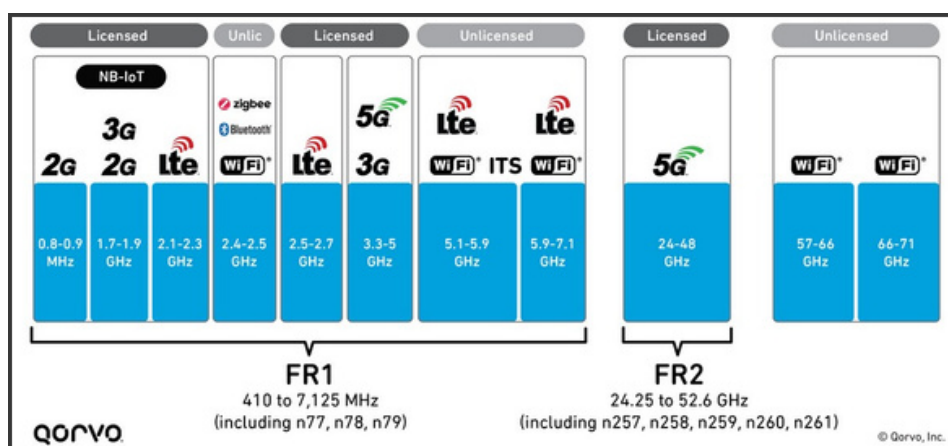


Figure 4. Frequency bands for major wireless standards.

Creating Thinner Layers

The ability to create thinner layers, with accurate control over thickness and uniformity, is critical to support higher frequencies. The layer thickness determines the frequency response of the filter: thickness is inversely proportional to frequency. At high frequencies, the layers must be extremely thin. Small variations in thickness have a major impact on the filter's frequency response; variations of just a few nanometers will create an unacceptable deviation in the response. For example, one nanometer of tungsten on a 5 GHz filter results in a frequency shift of approximately 10 MHz. Qorvo's BAW engineers were able to solve these challenges, capitalizing on the experience from producing and selling more than 20 billion BAW filters over the last 10 years.

Minimizing Losses

Maintaining low resonator loss in acoustic filters becomes inherently more difficult at higher frequencies, for a variety of different reasons. First, material losses generally increase with higher frequencies. In addition, as the layers within the resonators become thinner to support higher-frequency operation, the resonators must become smaller in area to maintain the same capacitance. Unfortunately, this means that the ratio of resonator perimeter to area increases, which makes the resonator more prone to loss lateral through acoustic leakage. Mitigating above challenges requires careful engineering of the resonator structure and a clear understanding of the underlying physics. Qorvo's BAW engineers have addressed these hurdles, resulting in lower insertion losses and steeper filter skirts.

Increased Bandwidth

A key advantage of the new higher-frequency bands allocated for 5G is that they offer much more bandwidth than is available in lower-frequency spectrum. 5G uses that additional bandwidth to deliver higher data rates and increase network capacity. As a result, 5G BAW filters must not only be able to operate at higher frequencies than previous filter generations, they must also be able to support those frequencies with much greater bandwidth.

Increasing filter bandwidth requires enhanced piezo-electric coupling. The use of Scandium-doped AlN piezo layers has been key to overcoming this problem – however, filter manufacturing becomes more challenging in comparison to regular AlN.

Key factors that influence piezoelectric coupling in BAW resonators are the piezo material, piezo layer quality, electrode configuration, acoustic reflector and parasitics.

Attaining perfect deposition for the piezo layer is challenging. A highly tuned pulsed DC magnetron sputtering process has proven to be the best choice in attaining a suitable layer for quality high-volume manufacturing.

The electrode configuration also significantly affects coupling. The thickness of the electrodes relative to the piezo layer must be optimized to meet the filter's frequency and performance requirements. For each electrode material, there is an optimum thickness ratio that delivers the highest coupling coefficient (k_{2eff}).

Filters for 5G bands n77 (3.3-4.2 GHz), n78 (3.3-3.8 GHz) and n79 (4.4-5 GHz) require wide bandwidth and benefit greatly from optimized coupling. Enhanced coupling is also required when designing multiplexers for CA applications.

By using Sc-doped AlN, filter designers can increase coupling while keeping losses constant, as shown in Figure 5 below. Choosing the right percentage of Sc allows for, a good compromise between filter skirt steepness and bandwidth.

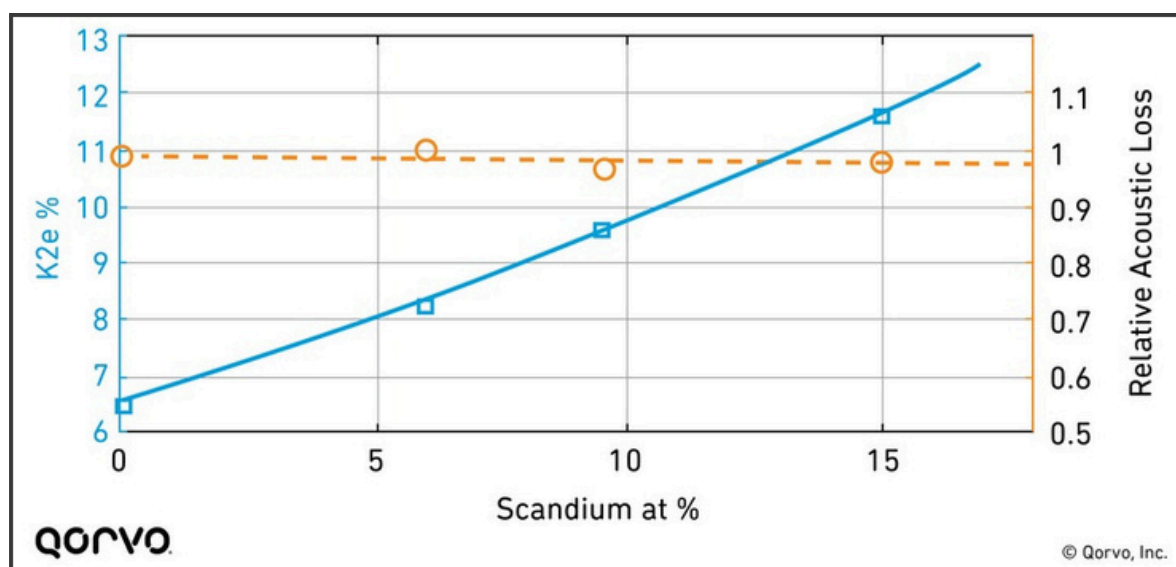


Figure 5. Measured coupling and relative acoustic loss of BAW resonators with various Sc%.

Thermal Issues and Reliability

5G is driving the need for smaller and sleeker devices. To maintain reliability in these devices, filters must be able to effectively dissipate the heat generated during operation. As BAW filters evolve to address higher frequencies, this becomes more challenging. Resonator areas shrink and layers become thinner, thus the overall resonator volume decreases. As a result, there is an increase in the power density in the device; the filter must dissipate heat more efficiently to maintain an acceptable operating temperature.

Fortunately, the structure of BAW helps to mitigate this problem, as shown in Figure 6. BAW filters dissipate heat to the substrate through the solid reflector layers. Because the layers are thinner at higher frequencies, this heat path becomes shorter, increasing the filter's ability to dissipate heat.

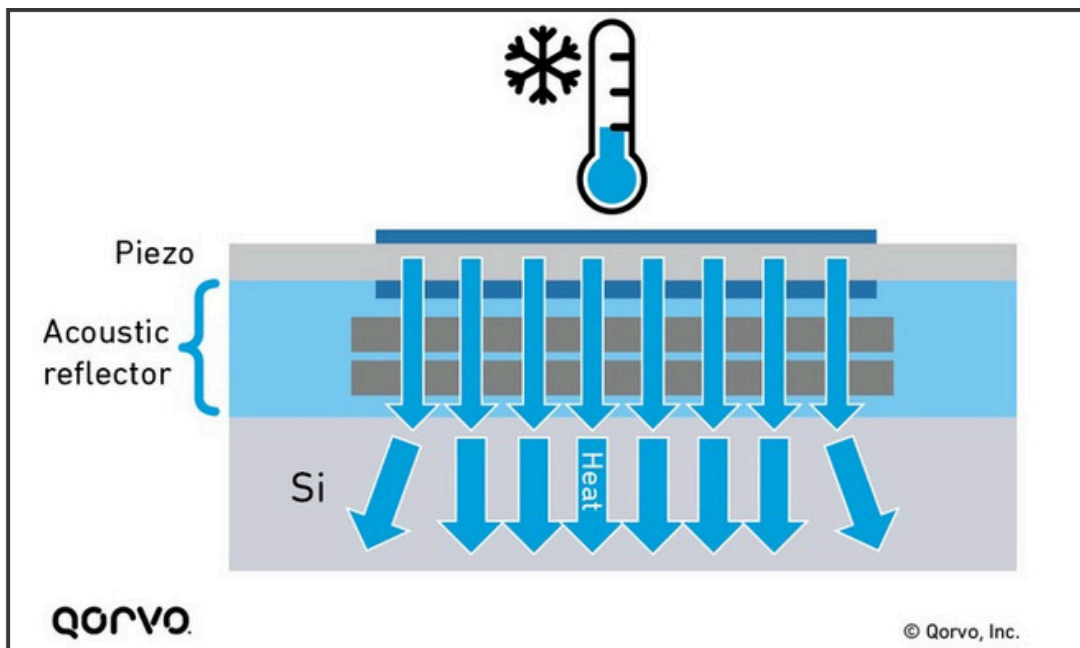


Figure 6. Heat dissipation path in BAW filters.

Reducing Size

RFFE complexity continues to increase rapidly, as shown in Figure 7 – 5G accelerates that process, due to the addition of new bands as well as new requirements such as 4x4 MIMO. At the same time, space allocated to the RFFE in handsets is actually shrinking as smartphone makers cram in more features that are desirable to consumers, such as bigger batteries and more-sophisticated cameras.

These trends mean there is constant pressure to make all RF components, including filters, smaller and more highly integrated. The growth of the IoT also adds to demand for very small filters and other RF components that can fit into tiny IoT devices.

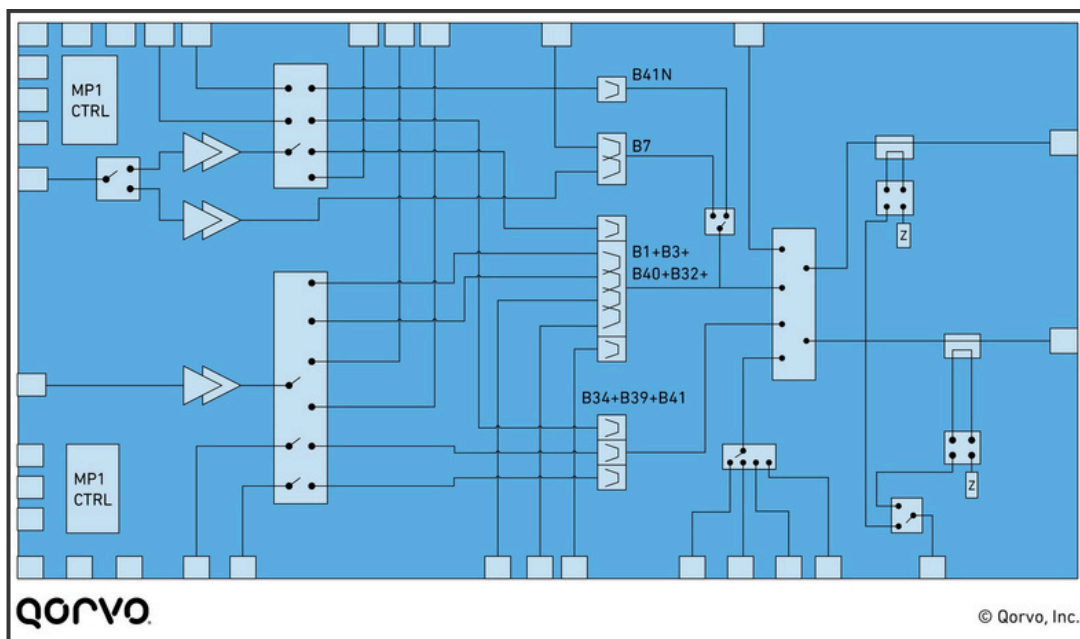


Figure 7. Mobile device front-end module with CA filter technology.

Qorvo's uBAW, shown in Figure 8, is a key step in a continuing strategy to reduce filter size. In traditional wafer level packaged (WLP) BAW filters, less than half of the total area of the filter is occupied by the resonators. The periphery of the filter, which includes the Cu-Sn input-output pillars, accounts for about 40% of the total area. uBAW almost eliminates this peripheral area entirely by placing the pillars on top of the WLP roof. This reduces the total size of the filter by an average of 30%. An example filter based on uBAW to reduce size is the Qorvo QPQ2200Q, a fully 50 Ohm matched automotive tested Wi-Fi and LTE coexistence filter with a footprint of 1.1x0.9x0.585 mm.

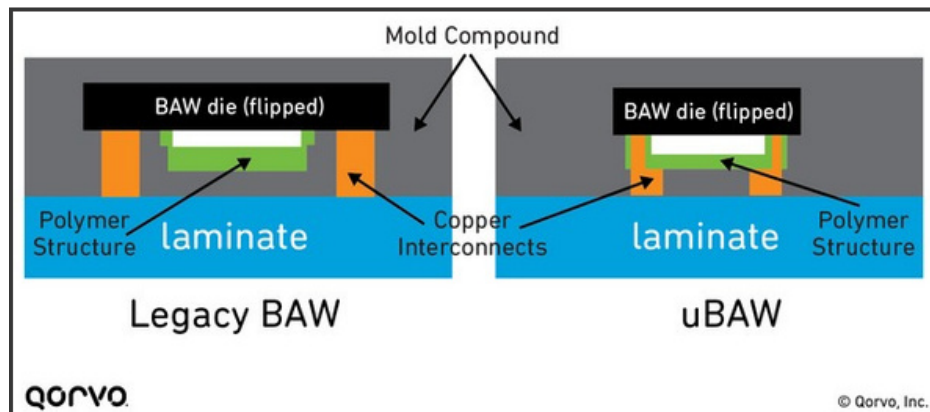


Figure 8. Simplified cross-section of legacy WLP BAW versus uBAW.

Complex Filters: Multiplexers and Antennaplexers

Complex modules that combine several filters have become important RFFE elements in recent years and play even bigger roles in 5G.

Multiplexers are employed to support carrier aggregation (CA). CA is used to deliver higher data rates by combining fragmented slivers of spectrum, often located in different frequency bands. Multiplexers include all the filters required to support uplink and downlink communications on each of the aggregated carriers, allowing concurrent operation on all of them while preventing interference between them. Considerable care and knowledge are required to design filters that work well together in a multiplexer, since the performance of each filter is impacted by other filters in the module. Unless the filters are carefully matched, a multiplexer may not perform adequately in terms of insertion loss and isolation. For example, signals could leak from a transmit to a receive path, desensitizing the receiver.

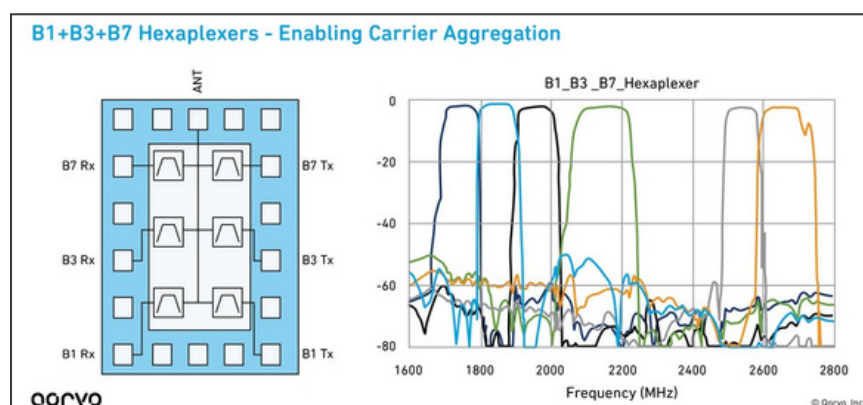


Figure 9. Band 1, 3, 7 multiplexer device.

Antennaplexers address another growing challenge: the limitation on the number of antennas that handsets can support. Trends such as the growing number of bands in a small package or device, 4x4 MIMO and the addition of standards like ultra-wide band (UWB) add to the pressure to increase the number of antennas in each handset. The problem is that handset size and space limitations make it hard to add yet more antennas while maintaining adequate performance. Antennaplexers reduce the need to add antennas by allowing more frequency bands and standards (such as 5G cellular bands, Wi-Fi, and GPS) to share each antenna. This frees more space for manufacturers to add features that are attractive to consumers, such as additional cameras and edge-to-edge screens. BAW antennaplexers offer low insertion loss, as well as excellent isolation and power handling, maximizing radiated power and reducing the risk of system de-sense.

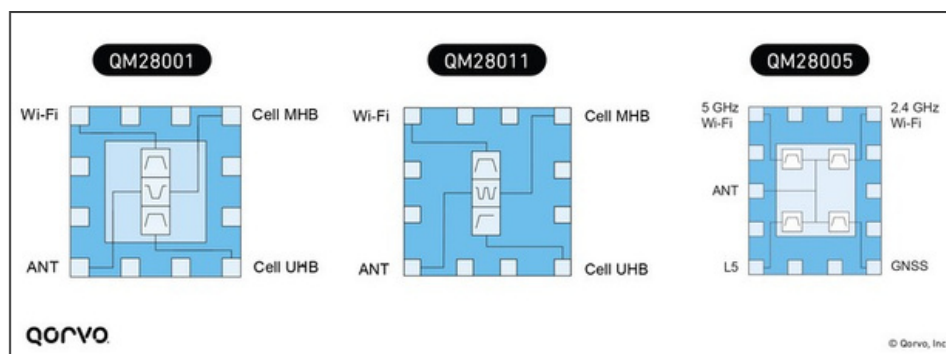


Figure 10. Examples of Qorvo antennaplexer devices.

Simulation

Optimization tools are ubiquitous in electronic design. BAW filter design optimization requires a scalable model that precisely predicts performance over a wide frequency range. The rapidly increasing complexity of RFFE architectures requires an unprecedented level of accuracy in simulation and modeling. The time allotted to

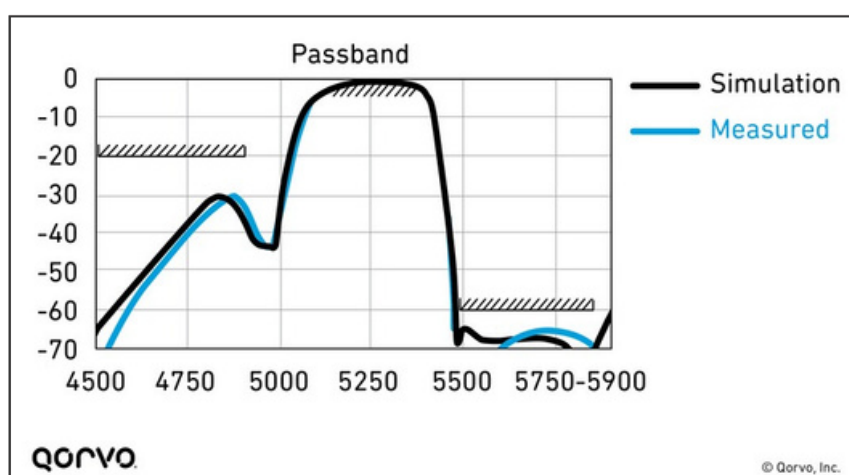


Figure 11. Comparison between simulation and measured data for SMR BAW.

develop a complex integrated RF module, from defining the required functionality to producing a final qualified sample, is very short and leaves no room for trial and error. Therefore, it is vital to be able to have extremely accurate methods for simulating the performance of each functional block during design, to provide confidence that the

final product will match the performance of the model.

Conclusion

Qorvo's BAW filter technology has been advancing to provide a formidable solution to the filtering challenges created by the global rollout of 5G. BAW filters are evolving to support higher frequencies and increased bandwidth with ever increasing performance. They are becoming smaller, more highly integrated and they are being incorporated into a growing range of complex multiplexers and antennaplexers. Highly accurate simulation methods are facilitating the rapid development of new filters and modules to meet stringent requirements. BAW technology will continue to evolve to address the future challenges presented by 5G and other wireless standards.

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Conclusion

This concludes the '2024 RF Filter Digest' eBook from everything RF. This comprehensive resource has been developed for those looking to learn about the current-generation RF Filters. This eBook discusses the challenges involved with 5G mmWave filtering, high-rejection LTCC filter performance, the role of MMIC filters in developing next-generation systems, and the impact of BAW filters on 5G applications. It also discusses filter design in the Cadence AWR design environment.

With the introduction of new wireless technologies, the frequency spectrum is getting very crowded. To ensure these technologies can operate simultaneously without impacting each other, RF Filters play a very important role. RF Filters are crucial in optimizing signal transmission and reception, selectively allowing desired frequencies to pass through while attenuating unwanted signals.

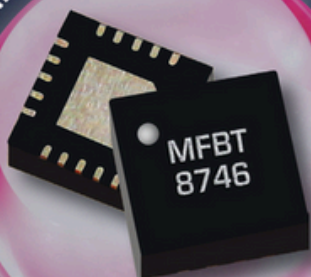
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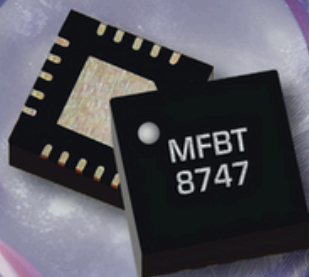


PERFORMANCE THAT SEEMS UNREAL

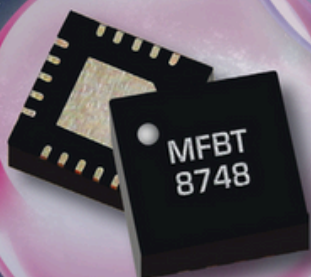
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